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Multi-objective Optimisation of Suction Cup and Parallel Jaw Grippers

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Abstract

Keywords: Pymoo, FastAPI

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Nomenclature

$\alpha_{\mathrm{EI}}(\mathbf{x})$ Expected improvement acquisition

β_{cov} Weight of the angular coverage term within the spatial diversity component

$\boldsymbol{\theta}$ Algorithm hyperparameters of the evolutionary algorithm

$\chi_{\mathcal{J}_j}$ Voxel occupancy grid of gripper jaw j

$\chi_{\mathcal{W}_i}$ Voxel occupancy grid of the workpiece in grasp orientation i

Δp Pressure difference

$\Delta\mathbf{R}_{ij}$ Relative rotation between the orientations of grasps i and j

$\Delta\mathrm{HV}_{\mathrm{tol}}$ Hypervolume-stagnation tolerance

δ_i Signed opening width of the placement selected for grasp γ_i

δ_{clr} Clearance threshold of the gripper–workpiece collision check

ℓ_{gn} Length of the spline-based gripping contour

γ_i Grasp i of the predefined grasp set

$\hat{\mathbf{u}}_i$ Normalised direction from the object centroid to grasp position i

$\hat{A}(\cdot)$ Normalised triangle-area objective of a gripper layout

$\hat{n}_{\mathrm{g}}(\cdot)$ Normalised grasp count of a gripper layout, i.e. the number of evaluated grasps divided by the clustering budget

$\kappa(\mathbf{x}, \mathbf{x}')$ Covariance/kernel function

λ Lebesgue measure

λ Weight of the spatial component in the total diversity score

$\mathbb{E}[\cdot]$ Expectation operator

$\mathbf{c}_{\mathrm{gr}}(\cdot)$ Complete gripper configuration generated from the decision vector

$\mathbf{e}$ Axis-aligned bounding-box side lengths of the workpiece

$\mathbf{e}$ Bounding-box side lengths

$\mathbf{f}(\cdot)$ Natural objective vector returned by the evaluation service

$\mathbf{f}(\mathbf{x})$ Objective function vector

$\mathbf{f}_{\mathrm{BO}}(\cdot)$ Objective observations supplied to Ax

$\mathbf{f}_{\mathrm{EA}}(\cdot)$ Minimisation-form objective vector supplied to the evolutionary algorithms

$\mathbf{F}_{\mathrm{EA}}^{(G)}$ Minimisation-form objective values of the population in generation G

$\mathbf{o}$ Centroid of the workpiece mesh

$\mathbf{P}$ Cartesian suction cup configuration derived from the reduced parameterization

$\mathbf{P}_i$ Spline interpolation point

$\mathbf{q}$ Decision vector representing a candidate gripper configuration

$\mathbf{r}$ Reference point

$\mathbf{R}_i$ Orientation matrix of grasp i

$\mathbf{R}_i$ Rotation matrix of the i-th grasp orientation

$\mathbf{s}_j$ Spline control point j of the jaw contour

$\mathbf{v}$ Relative displacement between the voxel grids

$\mathbf{x}$ Decision variable vector

$\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$ Dominance relation (solution a dominates b)

$\mathbf{x}_i$ Position of grasp i

$\mathbf{y}_i$ Position of the i-th grasp contact point

$\mathbf{y}_i$ Surface-projected position of grasp i used in diversity evaluation

$\mathbf{z}^{\mathrm{ref}}$ Reference point for hypervolume computation

$\mathcal{A}(\cdot,\cdot)$ Selected multi-objective evolutionary algorithm

$\mathcal{B}(\mathbb{R}^k)$ Borel σ-algebra on $\mathbb{R}^k$

$\mathcal{B}_{t+1}$ Batch of candidates generated by Bayesian optimization

$\mathcal{D}$	Observed data set
$\mathcal{D}_t$	Dataset of completed Bayesian optimization trials after t evaluations
$\mathcal{F}^{(G)}$	Objective vectors of the evolutionary population at generation G
$\mathcal{F}_{\mathrm{BO}}^{(t)}$	Non-dominated objective set after t completed Bayesian evaluations
$\mathcal{F}_{\mathrm{EA}}^{(G)}$	Non-dominated objective set after evolutionary generation G
$\mathcal{F}_{\sigma}$	Final objective vectors obtained with random seed σ
$\mathcal{GP}$	Gaussian process
$\mathcal{G}(\cdot)$	Gripper jaw geometry generated from the decision vector
$\mathcal{H}_{\mathrm{valid}}$	Set of valid grasps after seal-formation and collision filtering
$\mathcal{M}(\cdot)$	Bayesian surrogate-modeling procedure
$\mathcal{O}(kN^2)$	Computational complexity
$\mathcal{P}^*$	Set of Pareto-optimal solutions in the decision space
$\mathcal{Q}$	Admissible search space
$\mathcal{R}$	Set of predefined workpiece orientations of a two-finger optimization run
$\mathcal{S}(\cdot)$	Interpolating spline describing the jaw contact contour
$\mathcal{U}^{(G)}$	Evolutionary population of candidate decision vectors at generation G
$\mathcal{W}$	Workpiece mesh, with surface $\partial\mathcal{W}$

$\mathcal{W}_i$ Workpiece geometry in the orientation of grasp γ_i

$\mathcal{X}$ Feasible decision space

$\mathcal{Z}$ Set of random seeds used for meta-optimization

$\mathrm{HV}(\cdot,\cdot)$ Hypervolume indicator of an objective set with respect to a reference point

$\mathrm{HV}(A, \mathbf{r})$ Hypervolume indicator

$\mathrm{HV}^{(G)}$ Hypervolume indicator at evolutionary generation G

$\mathrm{HV}_{\mathrm{BO}}^{(t)}$ Bayesian hypervolume after t completed evaluations

$\mathrm{HV}_{\mathrm{EA}}^{(G)}$ Evolutionary hypervolume after generation G

$\mu(\mathbf{x})$ Predictive mean

DFT Discrete Fourier transform

$\Phi(\cdot)$ Transformation from reduced decision variables to Cartesian suction cup coordinates

$\psi(\cdot \mid \mathcal{D}_t)$ Bayesian acquisition function conditioned on the observed trials

$\rho_w(\cdot,\cdot)$ Combined pose distance with orientation weight w

$\rho_{\cos}(\cdot,\cdot)$ Cosine distance between the centroid-referenced directions of two grasps

$\rho_{\mathrm{geo}}(\cdot,\cdot)$ Geodesic distance between two grasp orientations

σ Random-seed index

$\sigma^2(\mathbf{x})$ Predictive variance

ε	Small numerical constant preventing division by zero
$\varepsilon_{\mathrm{col}}$	Numerical collision threshold of the FFT-based overlap map
$\widehat{\mathbf{f}}_t$	Probabilistic surrogate approximation of the objective vector after t evaluations
$\zeta_{i,j}$	Overlap map of jaw j and the workpiece for grasp γ_i
A	Approximation set
$A(\cdot)$	Area of the triangle formed by the three suction cups of a gripper layout
A_{seal}	Effective sealed area
$A_{\max}$	Maximum triangle area attainable within the admissible mounting circle
b	Maximum number of Bayesian candidates requested per batch
$B_{\max}$	Maximum number of Bayesian optimization batches
d	Number of decision variables
d	Number of decision variables
$D(\cdot)$	Diversity metric of the grasp distribution of a gripper layout
D_{cov}	Angular coverage of grasp positions around the object centroid
d_{geo}	Geodesic distance on $SO(3)$
D_{pos}	Normalized spatial diversity
D_{pos}	Spatial component of the diversity metric

D_{rot} Normalized orientational diversity

D_{rot} Orientation component of the diversity metric

D_{spread} Normalised mean pairwise distance between grasp positions

E Cumulative number of completed external objective evaluations

e_i i-th bounding-box side length

$E_{\max}$ Maximum number of evolutionary objective evaluations

$E_{\max}^{\text{BO}}$ Maximum requested number of Bayesian objective evaluations

$f_i(\mathbf{x})$ Individual objective function

F_{hold} Holding force

$f_{\min}$ Best observed objective value

f_{tol} Objective-space convergence tolerance

G Evolutionary generation index

$g_j(\mathbf{q})$ j-th inequality constraint

G_{final} Final evolutionary generation reached by the optimization

$G_{\max}$ Maximum number of evolutionary generations

h_j Spline depth parameter at relative contour position $j/4$, $j = 1,2,3$

k Number of objectives

K^*	Final number of clustered grasps selected by the elbow criterion
k_i	Proximity KPI `max_prox` of grasp γ_i reported to the optimizer
$k_i(\cdot)$	Grasp-quality KPI (`max_prox`) of grasp γ_i
K_{clst}	Clustering budget, i.e. maximum number of grasps retained by clustering
m	Number of objective functions
m	Number of spline interpolation points
$m(\mathbf{x})$	Mean function
N	Population size
n	Number of grasps in the considered set
n_c	Number of inequality constraints
n_{g}	Number of grasps used for diversity evaluation
n_{obs}	Number of observations
n_{or}	Number of predefined grasp orientations of a two-finger optimization run
N_{pop}	Population size of the evolutionary algorithms
n_{v}	Number of valid grasps returned by the feasibility filtering
n_{win}	Sliding-window length of the evolutionary termination criteria
p_i	Maximized jaw proximity score of grasp γ_i

p_{atm} Atmospheric pressure

$p_{i,x}$,$p_{i,y}$ Planar coordinates of suction cup i in the gripper frame

q_i^{L},q_i^{U} Lower and upper bounds of the i-th decision variable

q_{tol} Design-space convergence tolerance

r_{cup} Radius of a single suction cup

$r_{\min}$,$r_{\max}$ Lower and upper bound on the common suction cup radius

S Safety factor

s Characteristic object scale (bounding-box diagonal)

s Characteristic object scale derived from the mesh bounding box

$SO(3)$ Special orthogonal group of 3D rotations

t Number of completed Bayesian trials

w Orientation weight in the combined pose distance

$X_{\min}$,$X_{\max}$,$Y_{\min}$,$Y_{\max}$ Horizontal bounding-box coordinates of the workpiece

y_i Observed objective value at $\mathbf{x}_i$

z_0 Fixed height coordinate of the suction cups in the gripper frame

1 Introduction

1.1 Motivation

Robotic grasping is a fundamental capability in automated manufacturing, logistics, and service robotics[5]. The ability to reliably grasp objects of varying shapes, sizes, and surface properties remains a challenging problem, particularly in unstructured environments[6]. Industrial applications such as bin picking, assembly, and material handling require grippers that can operate robustly across a wide range of workpieces[7].

Among the commonly used end-effectors, suction-cup grippers and two-finger grippers represent two complementary approaches[8]. Suction grippers are widely used for handling objects with smooth surfaces due to their simplicity and speed, whereas two-finger grippers provide flexibility in handling objects with complex geometries. Designing optimal configurations for these grippers is non-trivial, as grasp performance depends on multiple interacting factors such as geometry, contact distribution, force limits, and collision constraints[9].

While the gripper configuration problems are well-motivated in themselves, the primary objective of this thesis is not simply to find the best gripper configuration, but to systematically apply and compare optimization algorithms for solving these configuration problems. A large number of optimization algorithms exist, spanning evolutionary strategies, decomposition-based methods, indicator-based selection, and Bayesian optimization, yet their relative suitability for gripper design problems of this type is not well understood. Different algorithms may exhibit substantially different convergence behaviour, solution quality, and robustness depending on the structure of the objective landscape and the dimensionality of the search space. Identifying which algorithm class performs best for each gripper problem is therefore a practically relevant and scientifically meaningful research question[10].

1.2 Problem Definition

The design of task-specific robotic grippers remains largely a manual and experience-driven process. Geometric parameters, such as the placement of suction cups or the contour of gripper jaws, are commonly selected iteratively and subsequently validated through simulation, geometric evaluation, or physical experiments [11]. This procedure can become time-consuming when many possible configurations must be considered.

For the gripper-design problems investigated in this thesis, the performance of a candidate configuration is determined by computational evaluation routines. These routines do not provide analytical gradients or closed-form expressions that directly relate the design parameters to the resulting performance. Furthermore, a gripper design must generally satisfy several criteria simultaneously, for example by performing across multiple workpieces, grasp locations, or geometric requirements. Improvements with respect to one criterion may reduce performance with respect to another. Representing all criteria by a single aggregated value could hide these trade-offs.

These characteristics lead to expensive, derivative-free, multi-objective black-box optimization problems [12]. A variety of optimization approaches can be applied to such problems, including evolutionary multi-objective algorithms and Bayesian optimization methods. However, their performance depends on factors such as the problem representation, number of objectives, objective landscape, and available evaluation budget [13, 14]. Consequently, results obtained for generic test problems or other applications cannot necessarily be transferred directly to task-specific gripper design.

The primary goal of this thesis is therefore to apply and comparatively evaluate selected evolutionary and Bayesian multi-objective optimization methods for automated gripper configuration. Two application cases are considered: the configuration of a multi-suction-cup gripper and the optimization of rigid jaw contours for a two-finger gripper. In the first case, the positions of the suction cups are optimized to obtain layouts that perform well for a given workpiece. In the second case, spline-based jaw contours are optimized to provide suitable geometric conformity at predefined grasp points. The two cases provide different geometric representations and objective structures through which the behaviour of the selected optimization algorithms can be investigated.

The objective is not to identify a universally superior optimization algorithm. Instead, the thesis examines how the selected methods perform on the two specific gripper-design

problems under comparable experimental conditions. The comparison considers the quality of the obtained multi-objective solutions, their development over the available evaluation budget, and the consistency of the results across repeated runs.

To achieve this goal, the following tasks are defined:

- **Problem formulation:** Formulate the multi-suction and two-finger gripper configuration problems as multi-objective black-box optimization tasks by specifying their decision variables, objectives, and applicable constraints.
- **Framework integration:** Establish a common optimization interface through which the selected algorithms can access both gripper-evaluation pipelines under consistent conditions.
- **Evaluation preparation:** Integrate and adapt the problem-specific evaluation procedures required to assess candidate suction-cup layouts and rigid jaw contours.
- **Experimental design:** Define representative workpieces, predefined evaluation configurations, evaluation budgets, initialization conditions, repeated runs, and performance indicators for a controlled and reproducible comparison.
- **Algorithm application and comparison:** Apply selected evolutionary and Bayesian multi-objective optimization methods to both gripper-design problems and compare their performance using suitable multi-objective indicators.
- **Result analysis:** Analyze the obtained results with respect to solution quality, convergence behaviour, robustness across repeated runs, and sensitivity to the characteristics of the investigated gripper-design problems.

2 Literature Review

2.1 Theoretical Background

2.1.1 Robotics and Object Handling

Robotics

An industrial robot is a automatically controlled, reprogrammable multipurpose manipulator(s), programmable in three or more axes, which can be either fixed in place or fixed to a mobile platform for use in automation applications in an industrial environment[15]. A typical robotic system consists of a kinematic chain of rigid links connected by actuated joints, a set of sensors, a controller and an end-effector mounted at the flange of the last link. The configuration of the manipulator is described by its joint variables, while the position and orientation of the end-effector are described in a Cartesian task space[16]. The mapping between these two spaces is established through forward and inverse kinematics, which form the basis for trajectory planning and motion control[17].

Modern manufacturing systems increasingly require flexibility to accommodate changing production demands, mass customization, increasing product variants, and decreasing batch sizes [18, 19]. Robot-cell reconfiguration and programming nevertheless remain time-consuming and often require specialist expertise [18]. Task-specific assets such as grippers, sensors, and actuators provide robot cells with the capabilities required for particular manufacturing tasks [18].

Bin Picking Problem

Bin picking describes the task of removing individual workpieces from an unordered accumulation of parts, typically supplied in a container, and transferring them to a defined target pose[20, 21]. The task combines perception, grasp planning, collision-free motion planning and physical grasp execution. Because the parts are randomly oriented and may be partially occluded or entangled, the pose of the next object to be grasped is not known in advance and must be estimated from sensor data at runtime[22, 23].

For gripper design, the decisive consequence of this setting is that a grasp cannot be planned for one privileged object pose[20]. The gripper must be able to reach and hold the workpiece across a broad range of orientations and partial accessibility conditions, which implies that a useful gripper configuration provides several alternative grasp candidates distributed over the object surface rather than one single optimal contact[20, 24]. Additional constraints arise from the confined working space inside the bin, where collisions between the gripper body and the bin walls or neighbouring parts restrict the set of feasible approach directions[23, 25]. These requirements directly motivate the objectives used in this thesis, in particular the diversity of the available grasps for the suction cup gripper and the geometric conformance of the finger contour for the two-finger gripper.

2.1.2 Gripper Theory and Geometric Representation

Grasp Theory

A grasp is a configuration of contact points between a gripper and a workpiece such that the gripper can resist external forces and torques applied to the object[26]. The classical criterion for grasp quality is *force closure*. A grasp is force-closed if the contact forces available at the contact points can produce any net wrench on the object, meaning that arbitrary disturbances can be resisted[27]. Force closure requires a sufficient number of contacts with appropriate friction properties and geometric placement.

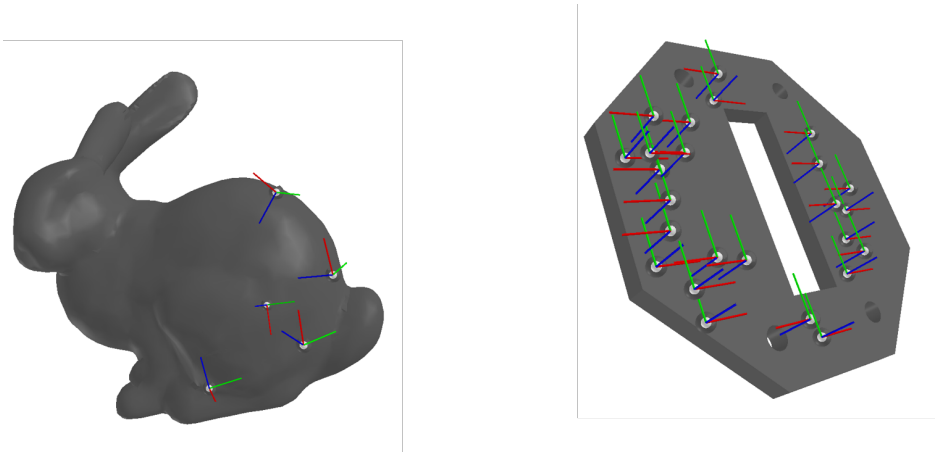


Figure 2.1: Visualisation of a grasp configuration on workpieces.

Figure 2.1 illustrates grasp configurations for two workpieces: the Stanford Bunny and the IPA Intermediate Plate. Each point on the workpiece surface encodes the translation and rotation required to transform the gripper's Tool Center Point (TCP) from its initial pose to the corresponding grasp pose.

Suction Cup Grippers

Suction cup grippers belong to the astrictive gripper class and generate holding force from a pressure difference between the evacuated volume inside the cup and the ambient atmosphere. The holding force is defined

$$F_{\text{hold}} = \frac{\Delta p \cdot A_{\text{seal}}}{S}, \qquad \Delta p \in (0, p_{\text{atm}}] \subset \mathbb{R}_{>0}, \quad A_{\text{seal}} \in \mathbb{R}_{>0}, \quad S \in [1, \infty) \subset \mathbb{R}, \quad F_{\text{hold}} \in \mathbb{R}_{>0} \tag{2.1}$$

where Δp is the pressure difference between the evacuated cup volume and the ambient atmosphere, bounded above by the atmospheric pressure p_{atm} since a perfect vacuum cannot be exceeded, A_{seal} is the effective sealed area enclosed by the cup lip, and $S \geq 1$ is a safety factor accounting for acceleration, orientation and friction conditions[28, 29]. The holding force is strictly positive for any non-vanishing pressure difference acting on a non-degenerate sealed area. The decisive requirement is therefore an airtight seal between the cup lip and

the workpiece surface, which restricts applicable surfaces to regions that are sufficiently smooth, non-porous and locally flat or only moderately curved relative to the compliance of the cup.

Practical implementations frequently use several cups arranged on a common mounting plate. Distributing the load over multiple cups increases the total holding force, provides redundancy if individual cups fail to seal, and, most importantly for handling, generates a resistance to tilting moments[30]. The moment capacity depends on the spatial arrangement of the active cups, which is commonly characterised through the area of the polygon or triangle spanned by the sealing contacts[29]. A large spanned area yields a stable grasp, whereas narrowly clustered cups behave similarly to a single contact. The placement of the cups on the mounting plate is thus the central design variable, and it must be chosen such that valid sealing contacts are achievable on every workpiece of the considered family.

Grasp Diversity

The *diversity* of a grasp set can be quantified by combining two complementary measures of spatial spread.

Geodesic Distance as a Diversity Measure. Orientational diversity is measured by calculating the pairwise geodesic distances between grasp orientations represented by rotation matrices. The geodesic distance describes the minimum angular rotation required to transform one orientation into another, accounting for the geometry of the rotation space. The mean of these pairwise angular distances is normalized by the maximum possible angular distance of 180°, resulting in a diversity score between zero and one. A score close to zero indicates that the grasp orientations are similar, whereas larger values indicate greater average angular separation and thus higher orientational diversity. This measure enables the assessment of how broadly candidate grasps cover different orientations, supporting the selection of diverse grasp configurations.

$$D_{\mathrm{rot}}(\{\mathbf{R}_i\}_{i=1}^{n}) = \frac{1}{180^\circ}\,\frac{1}{n(n-1)}\sum_{\substack{i,j=1\\ i\neq j}}^{n} d_{\mathrm{geo}}(\mathbf{R}_i,\mathbf{R}_j), \tag{2.2}$$

$$\mathbf{R}_i \in SO(3), \quad d_{\mathrm{geo}} : SO(3)\times SO(3) \to [0^\circ, 180^\circ], \quad D_{\mathrm{rot}} \in [0,1]$$

where $\mathbf{R}_i$ is the rotation matrix of the i-th grasp orientation and d_{geo} the geodesic distance on $SO(3)$, i.e. the minimum angular rotation required to transform one orientation into the other. Normalization by the maximum possible angular distance of 180° yields a score bounded by [0,1], where values close to zero indicate near-identical orientations and larger values indicate greater average angular separation[31, 32].

Spatial Diversity via Mean Pairwise Euclidean Distance. The spatial spread of a set of grasps can be quantified using the mean pairwise Euclidean distance between grasp positions in three-dimensional space. To ensure comparability across objects of different sizes, this distance is normalized by a characteristic object scale s. We define s as the Euclidean norm of the axis-aligned bounding-box side lengths:

$$s = \|\mathbf{e}\|_2, \qquad \mathbf{e} \in \mathbb{R}^3_{>0}, \quad s \in \mathbb{R}_{>0} \tag{2.3}$$

where $\mathbf{e}$ contains the side lengths of the bounding box. The normalized spatial diversity is then defined as

$$D_{\text{pos}}(\{\mathbf{y}_i\}_{i=1}^n) = \frac{1}{s}\,\frac{1}{n(n-1)} \sum_{\substack{i,j=1 \\ i\neq j}}^{n} \|\mathbf{y}_i - \mathbf{y}_j\|_2, \tag{2.4}$$

$$\mathbf{y}_i \in \mathbb{R}^3, \quad i,j \in \{1,\ldots,n\}, \quad n \in \mathbb{N}, n \geq 2, \quad s \in \mathbb{R}_{>0}, \quad D_{\text{pos}} \in [0,1]$$

where $\mathbf{y}_i \in \mathbb{R}^3$ denotes the position of the i-th grasp contact point and $n \geq 2$ is the number of grasps considered. Normalization by s makes the metric scale-invariant, allowing diversity values to be compared across workpieces of different sizes. Because every pairwise distance is bounded by the bounding-box diagonal, $\|\mathbf{y}_i - \mathbf{y}_j\|_2 \leq s$, the arithmetic mean of these distances is likewise bounded by s and therefore $D_{\text{pos}} \in [0,1]$. The lower limit is attained if and only if all grasp positions coincide, while the upper limit requires every pair of grasps to lie at diagonal distance and is thus only reachable for $n = 2$ at opposite corners of the bounding box. For $n \geq 3$ the attainable values are strictly smaller, so even well-distributed grasp sets are not expected to approach unity. This measure complements the geodesic measure by capturing absolute spatial separation rather than surface-constrained distance.

Multi Finger Grippers

Multi finger grippers are impactive grippers that transmit forces through direct contact between movable jaws and the workpiece. They are classified by the number of fingers, the kinematics of the jaw motion and the number of actuated degrees of freedom[33]. Two-finger parallel grippers with a single actuator represent the industrial standard because of their mechanical simplicity, robustness and low control effort[34]. Three-finger centric designs provide self-centring behaviour for rotationally symmetric parts, while anthropomorphic hands with many degrees of freedom offer high versatility at the cost of complexity.

Two grasp modes are commonly distinguished. Force-closure grasps hold the object through friction at the contact surfaces and therefore depend on the applied clamping force and the friction coefficient. Form-closure grasps immobilise the object through the geometry of the contacts alone, so that no disturbance can move the object without deformation[35]. Because a mechanically simple gripper provides few actuated degrees of freedom, the ability to realise form closure is shifted from the kinematics to the geometry of the jaws. This is the reason why application-specific jaw inserts, frequently produced additively, are widely used in industry and why the jaw contour is the natural design variable for optimization[36].

2.1.3 Optimization Theory

Black-Box Optimization

A black-box optimization problem is one in which the objective function has no known analytical form and no gradient information is available to the optimizer[37]. The function can only be queried at candidate input points, returning a scalar or vector output. This setting arises naturally in engineering design when objectives are evaluated through computationally expensive procedures, making gradient computation either impossible or impractical. Because each function evaluation may be costly, black-box optimizers must find high-quality solutions within a limited evaluation budget, placing emphasis on sample efficiency[38]. The gripper configuration problems addressed in this thesis fall squarely within this category. Objective values are computed through a series of geometric and numerical calculations applied to the candidate gripper configuration, and no closed-form expression exists that directly maps the decision variables to either the contact quality score or the grasp diversity metric.

Multi-Objective Optimization

Multi-objective optimization (MOO) refers to the simultaneous optimization of two or more conflicting objective functions over a shared decision space[39]. Formally, a multi-objective minimization problem is defined as

$$\begin{gathered}\min_{\mathbf{x}\in\mathcal{X}} \mathbf{f}(\mathbf{x}) = \big(f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_k(\mathbf{x})\big), \\ \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d, \quad d \in \mathbb{N}, \quad f_i : \mathcal{X} \to \mathbb{R}, \quad i \in \{1,\ldots,k\}, \quad k \in \mathbb{N}, k \geq 2, \quad \mathbf{f}(\mathbf{x}) \in \mathbb{R}^k\end{gathered} \tag{2.5}$$

where $\mathbf{x}$ is the decision variable vector, $\mathcal{X}$ is the feasible decision space, and $k \geq 2$ is the number of objectives. When objectives conflict, no single solution minimizes all objectives simultaneously. Instead, a set of trade-off solutions exists, collectively forming the Pareto front. The goal of a multi-objective optimizer is therefore not to find a single best solution but to approximate this trade-off surface as accurately and completely as possible[39, 40].

Pareto Optimality

The concept of Pareto optimality provides the theoretical foundation for comparing solutions in a multi-objective setting. A solution $\mathbf{x}^{(a)}$ is said to *dominate* $\mathbf{x}^{(b)}$, written $\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$, if and only if

$$\begin{gathered}f_i(\mathbf{x}^{(a)}) \leq f_i(\mathbf{x}^{(b)}) \quad \forall\, i \in \{1,\ldots,k\} \quad \text{and} \quad f_j(\mathbf{x}^{(a)}) < f_j(\mathbf{x}^{(b)}) \quad \text{for at least one } j \in \{1,\ldots,k\}, \\ \mathbf{x}^{(a)}, \mathbf{x}^{(b)} \in \mathcal{X}, \quad k \in \mathbb{N}, k \geq 2\end{gathered} \tag{2.6}$$

A solution is called *Pareto optimal* if no other feasible solution dominates it. The set of all Pareto-optimal solutions in the decision space is the *Pareto set*, and its image in the objective space is the *Pareto front*. In practice, algorithms compute an approximation of the Pareto front, and the quality of this approximation is assessed using performance indicators such as the hypervolume indicator[41, 42].

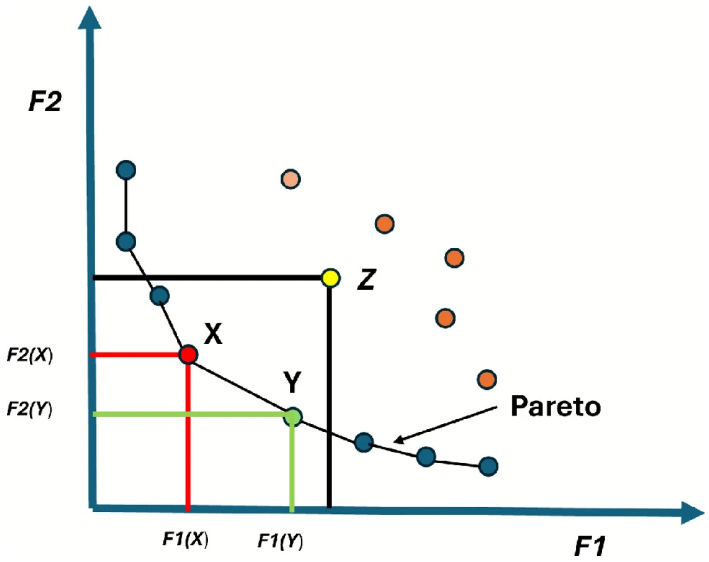


Figure 2.2: Example of a Pareto Front.[1]

Figure 2.2 illustrates the concept of dominated and non-dominated solutions in a multi-objective optimization (MOP) problem with two objectives. The objective is to minimize both objective functions. The connected curve represents the Pareto front (PF), which includes all Pareto-optimal solutions. Solution X_2 has higher values for both f_1 and f_2 when compared to solutions X_1 and Y. Thus, solutions X_1 and Y dominate solution X_2. However, since neither X_1 nor Y dominates the other, they are considered non-dominated solutions[1].

Many-Objective Optimization

Problems with four or more objectives are commonly referred to as many-objective optimization problems[43]. As the number of objectives grows, the proportion of mutually non-dominated solutions in a randomly sampled population increases rapidly, an effect known as dominance resistance. Pareto dominance then loses its ability to discriminate between candidates, so selection pressure towards the Pareto front deteriorates and the search degenerates into an essentially random walk guided mainly by diversity preservation[43, 44].

The number of solutions required to cover a k-dimensional front grows exponentially with k, visual inspection of the approximation set becomes impractical, and the computation of

indicators such as the hypervolume becomes expensive[4]. Algorithm design responds to these issues either by replacing dominance with decomposition into scalar subproblems, by introducing reference points or reference directions to structure the objective space, or by using an indicator as the selection criterion[43, 45].

No Free Lunch Theory

The No Free Lunch (NFL) theorems for optimization state that, averaged over the set of all possible objective functions, all non-repeating search algorithms exhibit identical expected performance[46, 47]. Any advantage that an algorithm demonstrates on one class of problems is therefore compensated by a corresponding disadvantage on another class. Corresponding results have been derived for the multi-objective case[47].

The practical consequence is that there is no universally superior optimization algorithm and that the question of which algorithm to use is only meaningful with respect to a specific problem class[48]. This provides the methodological justification for the empirical approach taken in this thesis. Rather than seeking a generally best algorithm, the objective is to determine which algorithms perform well on the particular structure of gripper configuration problems, and how sensitive this ranking is to workpiece geometry and to the dimensionality of the objective space.

Evolutionary Algorithms

Evolutionary algorithms (EAs) are a family of population-based metaheuristics inspired by the principles of biological evolution. A population of candidate solutions is maintained across discrete generations. At each generation, new candidate solutions are generated through variation operators, typically crossover, which recombines pairs of existing solutions, and mutation, which introduces random perturbations, and a selection mechanism retains the most promising individuals for the next generation based on their objective values. This loop continues until a termination criterion is met, such as a maximum number of function evaluations or convergence of the population[49].

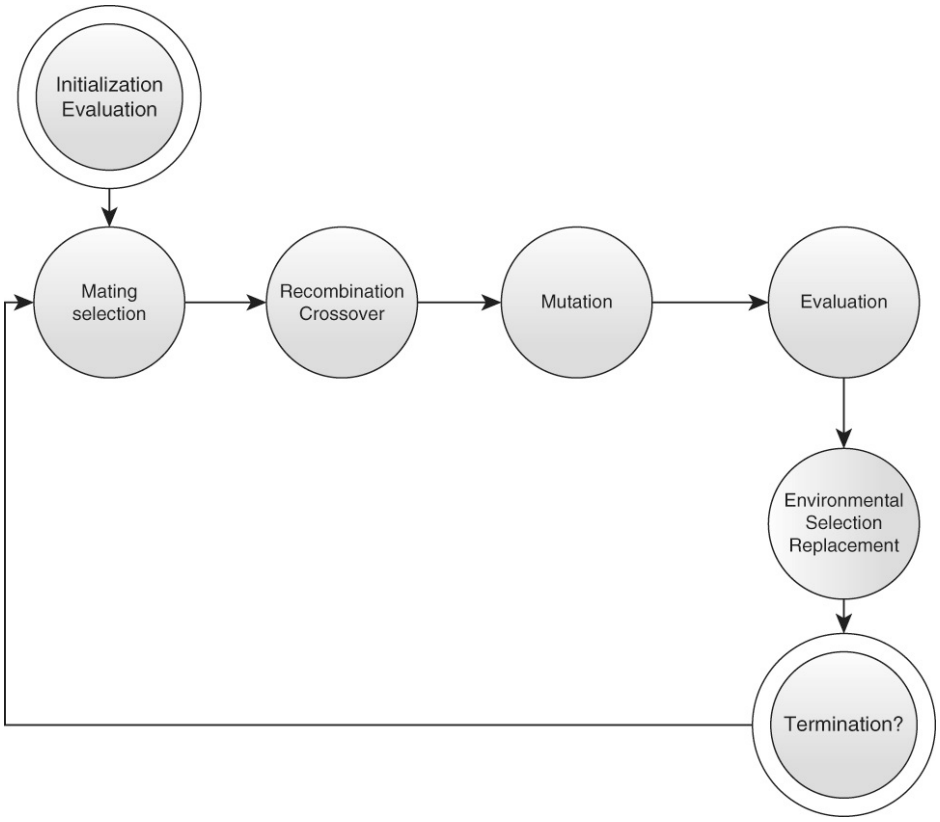


Figure 2.3: The evolutionary cycle, basic working scheme of all EAs [2].

Figure 2.3 shows the basic working scheme shared by all EAs. After initialization and evaluation of the initial population, the algorithm iteratively applies mating selection, recombination (crossover), mutation, evaluation of the offspring, and environmental selection (replacement), until a termination criterion is satisfied. The individual steps of this cycle can be described as follows.

Initialization and Evaluation An initial population of candidate solutions is created, usually by random sampling within the bounds of the decision space, so that the starting set covers the search space as broadly as possible without relying on prior knowledge. Every individual of this initial population is then evaluated by the objective function, which assigns each candidate the objective values on which all subsequent selection decisions are based. Since no variation operator has been applied yet, this population forms the reference state of generation zero.

Mating Selection Mating selection determines which individuals of the current population are allowed to produce offspring. Individuals with better objective values are assigned a higher probability of being selected, for example by tournament selection, which creates the selection pressure that drives the population toward better regions of the search space. The selection is stochastic rather than strictly deterministic, so that less competitive individuals retain a chance of reproducing and the genetic diversity of the population is preserved.

Recombination (Crossover) Recombination combines the decision variables of two or more selected parents into new offspring, so that building blocks which have proven advantageous in different individuals can be merged into a single solution. For continuous decision variables this is typically realized by operators such as simulated binary crossover, which generates offspring in the vicinity of the parent values. Since crossover exploits information that is already contained in the population, it is primarily responsible for the exploitative behavior of the algorithm.

Mutation Mutation applies small random fluctuations to individual decision variables of the offspring and thereby introduces genetic material that is not present in the parent population. It enables the algorithm to escape local optima and counteracts the loss of diversity caused by repeated selection and recombination. Mutation probability and mutation strength control the balance between exploration and exploitation: large values lead to a more random search, whereas small values keep the offspring close to their parents.

Evaluation of the Offspring The newly generated offspring are evaluated by the objective function in the same way as the initial population. In the case of expensive black-box objectives, this step dominates the computational cost of the entire cycle, whereas the variation and selection operators themselves are computationally negligible. The number of objective evaluations is therefore commonly used as the budget measure of an evolutionary run.

Environmental Selection (Replacement) Environmental selection forms the population of the next generation from the union of parents and offspring, so that the population size remains constant. In the single-objective case this reduces to retaining the individuals with the best objective values, while in the multi-objective case the decision is based on Pareto dominance together with an additional diversity criterion. Because the best individuals found so far can

be carried over unchanged, this step is what renders the progress of the population across generations monotone.

Termination After environmental selection, a termination criterion is checked. If it is not satisfied, the cycle continues with the mating selection of the next generation, otherwise the current population, or the non-dominated subset of all evaluated solutions is returned as the result. Typical criteria are a maximum number of generations or objective evaluations, a limit on the computation time, or stagnation of the objective values or of a quality indicator over a number of consecutive generations.

EAs are well suited to black-box multi-objective optimization because they require no gradient information, naturally maintain a diverse population of trade-off solutions, and are robust to noisy or discontinuous objective landscapes. Multi-objective evolutionary algorithms (MOEAs) extend the basic EA framework with selection mechanisms based on Pareto dominance or decomposition, enabling them to approximate the Pareto front in a single run[50, 51].

Bayesian Optimization

Bayesian optimization (BO) is a sequential model-based strategy for the global optimization of expensive black-box functions. A probabilistic surrogate model, most commonly a Gaussian process (GP), is fitted to the observations collected so far[52]. The surrogate provides both a mean prediction and an uncertainty estimate at unobserved locations, which together define an *acquisition function* that balances exploration of uncertain regions with exploitation of promising ones[52, 53]. The point that maximizes the acquisition function is selected as the next evaluation, after which the surrogate is updated and the process repeats[54][3]. Because the surrogate concentrates the sampling effort on informative regions, BO typically requires far fewer function evaluations than population-based methods, which makes it attractive whenever the evaluation budget is the limiting factor.

Gaussian Process Surrogates. A Gaussian process defines a distribution over functions and is fully specified by a mean function $m(\mathbf{x})$ and a positive semi-definite covariance or kernel function $\kappa(\mathbf{x}, \mathbf{x}')$, written as $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}'))$. Conditioning the prior on the observed data set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_{\text{obs}}}$, where $y_i \in \mathbb{R}$ is the response measured at $\mathbf{x}_i$ and n_{obs}

the number of evaluations performed so far, yields a posterior that is again Gaussian, with closed-form predictive mean $\mu(\mathbf{x})$ and non-negative predictive variance $\sigma^2(\mathbf{x})$ at any query point[55]. The kernel encodes the assumed smoothness and length scale of the underlying function, with the Matérn and squared exponential kernels being the most common choices, and its hyperparameters are estimated by maximizing the marginal likelihood[55]. An additional noise term accounts for stochastic or numerically imprecise evaluations[56]. The main limitations of GP surrogates are the cubic scaling of exact inference with the number of observations and a loss of predictive quality in high-dimensional decision spaces[52].

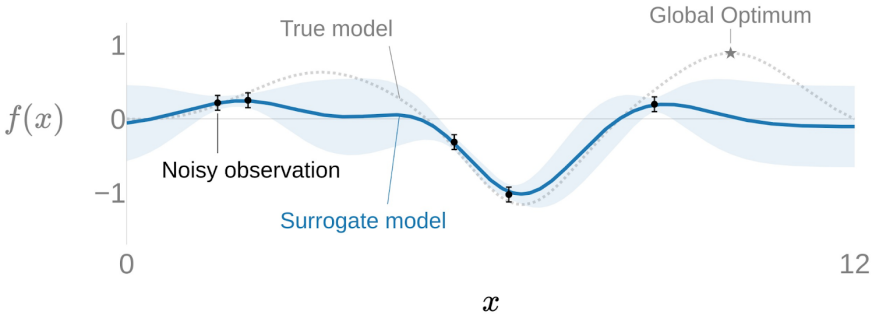


Figure 2.4: Surrogate Model [3].

In figure 2.4 a surrogate model is fit to five noisy observations using a GP to predict the objective, depicted by the solid line, and uncertainty estimates, illustrated by the width of the shaded bands. This objective is predicted for the entire range of possible parameter values, corresponding to the full x-axis. Importantly, the model is able to predict the outcome and quantify the uncertainty of configurations that have not yet been tested. Intuitively, the uncertainty bands are tight in regions that are well-explored and become wider as we move away from them.

Acquisition Functions. The acquisition function converts the posterior of the surrogate into a scalar utility that quantifies how promising a candidate point is. Expected improvement (EI) measures the expected amount by which a candidate improves on the best observed value,

$$\begin{aligned} \alpha_{\mathrm{EI}}(\mathbf{x}) &= \mathbb{E}\big[\max\big(f_{\min} - f(\mathbf{x}), 0\big)\big], \\ \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d, \quad f_{\min} &= \min_{i \in \{1,\ldots,n_{\mathrm{obs}}\}} y_i \in \mathbb{R}, \quad \alpha_{\mathrm{EI}} : \mathcal{X} \to \mathbb{R}_{\geq 0} \end{aligned} \tag{2.7}$$

where $\mathbb{E}[\cdot]$ denotes the expectation taken with respect to the posterior of the surrogate at $\mathbf{x}$ and $f_{\min}$ is the best objective value observed so far. Truncating the improvement at zero renders the acquisition non-negative, so that candidates which are not expected to improve on the incumbent receive zero utility. The expression admits a closed form for a Gaussian posterior[56, 3].

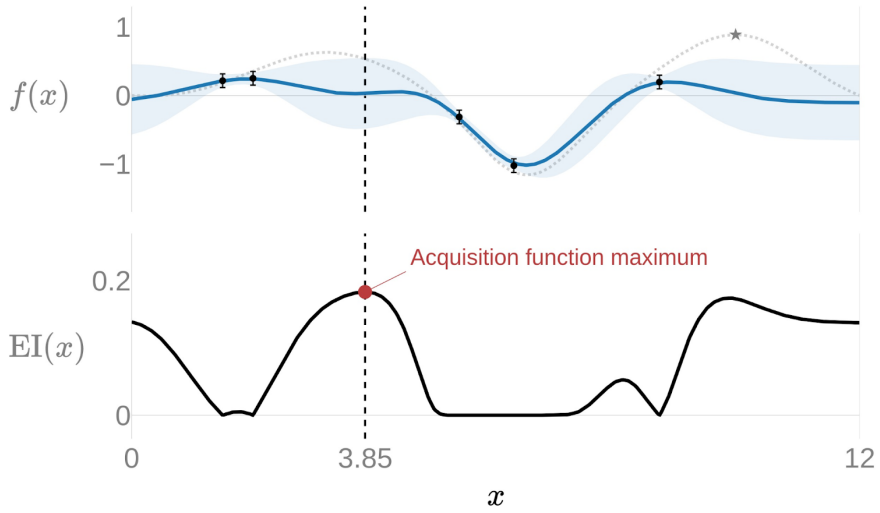


Figure 2.5: Acquisition Function [3].

A visualization of the expected improvement based on the surrogate model predictions is shown in Figure 2.5, where the next suggestion is where the expected improvement is at its maximum. Once a new highest EI is selected and evaluated, the surrogate model is retrained and a new suggestion is made. This process continues iteratively until a stopping condition, set by the user, is reached [3].

Since the acquisition function is inexpensive to evaluate and analytically differentiable, its maximization is performed by an inner optimizer rather than by querying the expensive objective.

Multi-Objective Bayesian Optimization. Extending BO to several objectives requires a utility notion for sets instead of single points. One family of methods fits an independent surrogate per objective and scalarizes the problem with randomly drawn weight vectors, so that the front is sampled gradually over successive iterations[57]. A second and now dominant family uses the hypervolume as the underlying utility[58]. Expected hypervolume improvement and its noisy, batch-capable variants select the candidates that are expected to enlarge the dominated hypervolume the most, which aligns the acquisition criterion directly with the indicator used for evaluation[57]. Information-theoretic approaches instead maximize the information gained about the Pareto front. Multi-objective BO is particularly appealing under tight evaluation budgets, but its cost per iteration grows with the number of objectives and observations, which makes the comparison against evolutionary methods on the gripper problems a question of the available budget rather than of asymptotic quality alone.

Performance Indicators

Because a multi-objective optimizer returns a set rather than a single point, comparing algorithms requires indicators that map an approximation set to a scalar quality value[4]. Indicators are usually grouped according to whether they assess convergence towards the front, the diversity of the solutions along the front, or both properties at once.

The *hypervolume* indicator (HV) measures the volume of the objective space that is dominated by the approximation set and bounded by a reference point $\mathbf{r}$,

$$\begin{gathered} \mathrm{HV}(A,\mathbf{r}) = \lambda\Big(\bigcup_{\mathbf{a}\in A} [\mathbf{f}(\mathbf{a}),\mathbf{r}]\Big), \\ A \subset \mathcal{X},\, |A| < \infty, \quad \mathbf{r} \in \mathbb{R}^k \text{ with } f_i(\mathbf{a}) \le r_i \;\forall\, \mathbf{a} \in A, i \in \{1,\dots,k\}, \\ \lambda : \mathcal{B}(\mathbb{R}^k) \to \mathbb{R}_{\ge 0} \text{ (Lebesgue measure)}, \quad \mathrm{HV} \in \mathbb{R}_{\ge 0} \end{gathered} \tag{2.8}$$

where λ denotes the Lebesgue measure on the Borel sets $\mathcal{B}(\mathbb{R}^k)$ and A is the finite approximation set returned by the optimizer[41, 59]. The reference point $\mathbf{r}$ must be dominated by every member of A, that is $f_i(\mathbf{a}) \le r_i$ for all $\mathbf{a} \in A$ and all i, so that each box $[\mathbf{f}(\mathbf{a}),\mathbf{r}]$ is non-empty and the indicator is well defined and non-negative. The hypervolume is the most widely used indicator because it is Pareto compliant, meaning that a strictly dominating

set never receives a worse value, and because it requires no knowledge of the true Pareto front[4, 41].

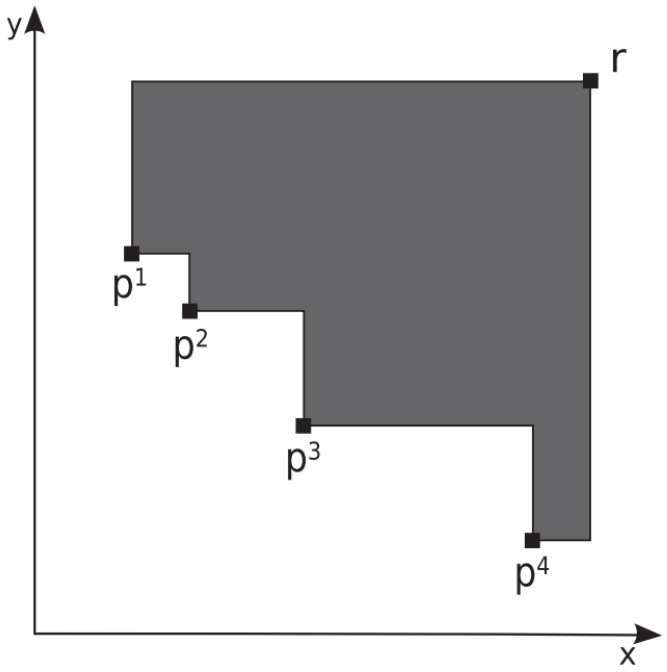


Figure 2.6: Hypervolume indicator [4].

Figure 2.6 shows a two-dimensional example of the hypervolume (an area) and a fixed reference point r.

Inverted generational distance. The *inverted generational distance* (IGD) measures the mean distance from each point of a reference set Z to the closest point of the approximation set and therefore requires a reference front, which is normally unavailable for real-world black-box problems[59]. IGD is furthermore not Pareto compliant: a set that dominates another can receive a worse indicator value[59]. The modified variant IGD^+ removes this defect by replacing the Euclidean distance with a dominance-aware distance that only penalizes those objective components in which the approximation set is actually worse than the reference point[60].

$$\mathrm{IGD}^{+}(A, Z) = \frac{1}{|Z|} \sum_{\mathbf{z} \in Z} \min_{\mathbf{a} \in A} \sqrt{\sum_{i=1}^{k} \big(\max\{\, a_i - z_i, 0 \,\}\big)^2}, \qquad \mathrm{IGD}^{+} \in \mathbb{R}_{\geq 0}, \tag{2.9}$$

where A is the approximation set under evaluation and a_i and z_i denote the i-th objective of an approximation and a reference point, respectively. Lower values indicate better approximation quality, and IGD^{+} is weakly Pareto compliant[60]. Because the true Pareto front of the gripper design problem is unknown, the reference set Z is constructed as the non-dominated union of the fronts of all compared runs[59]. IGD^{+} is therefore a *relative* measure: it quantifies how well an individual run approximates the best front jointly attainable by the whole set of runs, and it is optimistically biased in favour of those runs that contribute many points to Z[59].

2.1.4 Multi-Objective Evolutionary Algorithms

The candidate algorithms evaluated in this thesis are all members of the multi-objective evolutionary algorithm (MOEA) family. Each follows the general EA loop described in Section 2.1.3 but differs in how it guides selection toward a well-distributed approximation of the Pareto front. Brief introductions to each algorithm are given below.

NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is among the most widely adopted MOEAs. Its selection mechanism is based on two criteria applied in sequence. First, the entire population is sorted into non-domination ranks, so that all non-dominated solutions receive rank 1, they are removed, and the process repeats until every individual has a rank. Second, within the same rank, solutions are ordered by *crowding distance*, a density estimate that measures how isolated a solution is from its neighbours in the objective space. Solutions with a higher crowding distance are preferred, encouraging spread along the Pareto front. NSGA-II uses an elitist strategy in which the combined parent and offspring population of size $2N$ is truncated to N using these two criteria. Its computational complexity per generation is $\mathcal{O}(kN^2)$, where k is the number of objectives and N is the population size[61].

NSGA-III

NSGA-III extends NSGA-II to problems with three or more objectives (many-objective problems), where crowding distance becomes an unreliable diversity measure. Instead of crowding distance, NSGA-III uses a set of structured *reference points* distributed uniformly on a normalized hyperplane in the objective space. Solutions are associated with their closest reference point, and selection preference is given to solutions that are associated with underrepresented reference points and lie closest to their reference line. This mechanism explicitly promotes diversity across the objective space and has been shown to outperform NSGA-II on problems with three or more objectives[62].

MOEA/D

MOEA/D (Multi-Objective Evolutionary Algorithm based on Decomposition) takes a fundamentally different approach from Pareto-ranking methods. The multi-objective problem is decomposed into N scalar subproblems using a set of evenly distributed weight vectors and an aggregation function such as the weighted Chebyshev scalarization. Each subproblem is optimized simultaneously, with solutions sharing information within a defined neighbourhood of weight vectors. This neighbourhood-based collaborative search makes MOEA/D computationally efficient and well suited to problems where the Pareto front has a regular structure. Its complexity per generation is lower than NSGA-II, as it avoids global nondominated sorting[63, 64].

SMS-EMOA

The S-Metric Selection Evolutionary Multi-Objective Algorithm (SMS-EMOA) uses the hypervolume indicator directly as its selection criterion. It operates as a steady-state algorithm, so that at each step a single offspring is generated, added to the population, and the individual that contributes the least to the population hypervolume is removed. This ensures that the population always converges toward solutions that maximize the dominated hypervolume with respect to a reference point. SMS-EMOA is particularly well suited to problems where solution quality is to be measured by the hypervolume indicator, as its selection pressure is directly aligned with that metric[65].

CMOPSO

The Competitive Multi-Objective Particle Swarm Optimizer (CMOPSO) adapts the particle swarm optimization (PSO) framework to the multi-objective setting. In standard PSO, each particle moves through the decision space guided by its personal best position and a global best position. CMOPSO introduces a competitive mechanism in which particles compete in pairs within the swarm, and the losing particle updates its velocity by learning from the winning particle rather than from a fixed global best. An external archive of non-dominated solutions is maintained to provide guidance and preserve diversity across the Pareto front. This competitive selection strategy avoids premature convergence to a single region of the Pareto front and has been shown to achieve fast convergence while remaining competitive with dominance-based MOEAs on standard benchmark problems[66].

Suction Cup Grippers

2.2 State of the Art

This section compares previous studies according to their optimization target, the gripper properties treated as fixed, and their relevance to task-specific gripper configuration. The reviewed literature can be divided into four groups: grasp planning with fixed end-effectors, multi-suction grasping, two-finger gripper design, and comparative studies of multi-objective optimization algorithms.

2.2.1 Grasp Planning with Fixed End Effectors

Most robotic grasping research treats the end-effector geometry as a fixed input and searches for a feasible grasp pose on the workpiece. Zurbrügg et al. [67], for example, optimize grasp selection while assuming that the geometry and contact configuration of the gripper have already been defined. Analytical grasp-planning methods evaluate candidate contacts using geometric, force-closure, or form-closure conditions, whereas data-driven methods estimate grasp quality from training data. Despite their methodological differences, both approaches primarily vary the grasp pose rather than the gripper geometry.

Although Todescato et al. [5] establish gripper geometry as an optimization variable, their formulation is limited to a single suction cup. Their objective-function evaluation is methodologically similar to the approach used in this thesis, as suction-contact feasibility is assessed using quasi-static criteria related to the methods presented by Kleeberger et al. [68] and the compliant suction-contact model of Mahler et al. [69]. However, their formulation primarily considers the availability of valid suction grasps and does not explicitly evaluate the distribution/coverage of the resulting grasp set. The present thesis extends this approach to a multi-suction-cup layout by optimizing the spatial positions of the individual cups and evaluating valid grasps using orientation and spatial diversity metrics. These metrics distinguish between layouts that provide a similar number of valid grasps but differ in how broadly those grasps are distributed across positions and orientations.

The studies therefore address complementary but different questions. Grasp planning determines how an existing gripper should interact with a workpiece, whereas gripper-configuration optimization determines which geometry should be available before grasp planning is performed. Improving the grasp planner cannot fully compensate for a suction layout that cannot establish sufficient seals or for a jaw contour that cannot reach suitable contact regions. Consequently, fixed-gripper grasp-planning results do not directly resolve the end-effector design problem considered in this thesis.

2.2.2 Comparison of Multi-Suction Grasping Studies

Existing studies on multi-suction grippers address grasp prediction, experimental stability assessment, and load-aware motion planning, but generally assume predefined suction-cup layouts. Schillinger et al. [70] predict grasp quality from RGB-D images and optimize the grasp pose and gripper selection from a predefined set of multi-suction grippers. Their method selects among existing layouts rather than optimizing the positions of the individual cups.

Geng et al. [71] experimentally investigate how suction-cup size, geometric arrangement, centre of gravity offset, and robot acceleration influence grasp stability. Although different arrangements are evaluated, these are predefined experimental configurations rather than layouts generated through multi-objective optimization.

Lee et al. [72] model the load distribution among multiple suction cups using a quadratic program. They incorporate the resulting grasp-failure constraints into time-optimal trajec-

tory planning to support the fast and reliable manipulation of heavy objects. Their work focuses on load prediction and motion planning for an existing gripper configuration, not on optimizing the cup arrangement.

These studies therefore address complementary aspects of multi-suction grasping. Schillinger et al. focus on grasp-pose and gripper selection, Geng et al. examine the effects of gripper configuration and handling conditions experimentally, and Lee et al. model load distribution during dynamic manipulation. However, none of these studies formulates the positions of the individual suction cups as decision variables in a multi-objective layout-optimization problem.

2.2.3 Comparison of Two-Finger Gripper Studies

Research on two-finger gripper design includes the optimization of compliant fingers, underactuated mechanisms, and task-specific jaw geometries. Sun et al. [73] use three-dimensional topology optimization to design compliant continuum fingers and a lightweight gripper base. The resulting fingers deform passively to adapt to objects with different shapes. Similarly, Xie et al. [74] optimize the topology of flexible fingers while accounting for contact behaviour and contact-stress constraints. Their objective is to achieve adaptive, damage-free grasping of fragile and geometrically variable objects. Both studies focus on structural compliance and deformation, whereas this thesis considers the geometric optimization of rigid jaw contours.

Dong et al. [75] optimize the geometry and transmission system of an underactuated tendon-driven gripper. Their design variables include phalange dimensions, palm width, joint-mandrel radii, and tendon-routing parameters. The optimization aims to improve force transmission and grasp stability. Although this approach optimizes gripper geometry, it concerns the dimensions and actuation mechanism of an articulated gripper rather than the contact contour of a rigid parallel jaw.

Reichert [76] developed the direct predecessor of the parallel-jaw optimization framework continued in this thesis. The work represents the contact contour of a rigid parallel jaw using a parametric spline and evaluates candidate contours through a collision-map-based procedure. Intersections close to the intended workpiece boundary contribute positively to a geometric key performance indicator (KPI), whereas intersections extending into the workpiece interior are penalized. The KPI therefore measures the geometric conformity of

the generated jaw contour at a specified evaluation configuration rather than providing a complete force- or wrench-based assessment of grasp stability.

Reichert formulates the jaw-design problem using two fixed workpiece orientations, each contributing an objective to the optimization. This demonstrates the feasibility of combining spline-based jaw generation, geometric KPI evaluation, and multi-objective black-box optimization. However, the formulation is restricted to the two investigated evaluation configurations.

This thesis continues Reichert's work by extending the evaluation from two fixed configurations to a set of n predefined grasp points. The grasp points remain fixed and are not optimization variables; the decision variables are the parameters defining the spline-based jaw contour. Each candidate contour is evaluated at every predefined grasp point, producing one geometric KPI for each point. All n KPIs are retained as separate objectives and maximized simultaneously, resulting in an n-objective optimization problem with an n-dimensional objective space.

The optimization therefore seeks Pareto-optimal rigid jaw contours that provide different compromises in geometric conformity across the predefined grasp points. Compared with the studies on compliant and underactuated grippers, the formulation concerns rigid task-specific contact contours. Compared with Reichert's direct predecessor formulation, the principal extension is the evaluation and optimization of one separate objective for each of the n predefined grasp points.

2.2.4 Comparative Studies of Multi-Objective Optimization Algorithms

Comparative studies show that optimizer performance depends strongly on the problem and experimental setup. Tanabe and Ishibuchi [77] evaluate evolutionary algorithms on a suite of real-world multi-objective problems with varied characteristics, while Tian et al. [78] propose selecting algorithms based on problem features. Ishibuchi et al. [79] further demonstrate that termination conditions, population sizes, performance indicators, and test problems can change algorithm rankings. Audet et al. [41] emphasize that indicators assess different properties of Pareto-front approximations, such as convergence, distribution, and spread.

For expensive optimization, Qin et al. [80] compare two combinations of evolutionary and Bayesian optimization on ZDT and DTLZ problems and find that Bayesian evolutionary optimization is more efficient under their experimental conditions. Santoni et al. [13] compare high-dimensional Bayesian optimization methods on single-objective BBOB functions and report that performance varies with dimensionality, function landscape, and evaluation budget. These findings are therefore specific to the investigated problems and settings.

Such results cannot be transferred directly to the collision-aware suction-layout and spline-based jaw-contour problems considered in this thesis. Therefore, this thesis compares selected multi-objective evolutionary and Bayesian algorithms under common evaluation budgets, initialization conditions, repeated runs, and statistical procedures. The aim of this comparison is to identify methods suitable for these specific gripper-design problems rather than a universally superior optimizer.

2.2.5 Comparative Synthesis and Research Gap

The reviewed literature reveals three related research gaps. First, grasp-planning studies generally optimize grasp poses for predefined end-effectors rather than the gripper configuration itself. Second, existing multi-suction studies generally assume predefined cup layouts. For parallel-jaw design, Reichert [76] provides the spline-based formulation used in this thesis but considers only two fixed evaluation configurations. Third, existing algorithm comparisons do not determine which optimization methods are best suited to the two geometric gripper-design problems considered in this thesis.

For the multi-suction gripper, the unresolved problem is to optimize the positions of the individual cups to maximize grasp coverage on every workpiece and the area formed by the cup arrangement. For the two-finger gripper, this thesis continues Reichert's framework by optimizing a rigid spline-based jaw contour across a predefined grasp points. Each grasp point contributes an individual geometric conformity objective. Both formulations are expensive multi-objective black-box problems, but their different representations, numbers of objectives, and evaluation procedures may favour different optimization methods.

Within the literature reviewed for this thesis, no controlled comparison of evolutionary and Bayesian multi-objective algorithms was identified for these two gripper-design formulations.

2.2.6 Research Positioning and Contribution

This thesis addresses these gaps by treating gripper geometry as the primary optimization variable. The multi-suction formulation optimizes the spatial arrangement of individual suction cups, whereas the two-finger formulation optimizes spline-based contours for rigid parallel jaws. Candidate designs are assessed using problem-specific geometric and collision-aware evaluations.

A modular framework separates geometric evaluation from optimization through a common interface. This allows the applicable algorithms to operate on the same problem formulation, constraints, and evaluation conditions. Selected evolutionary and Bayesian multi-objective algorithms are compared under defined evaluation budgets using repeated independent runs, convergence histories, multi-objective performance indicators, and statistical tests. Evaluation counts and optimizer overhead are reported separately where computational cost is relevant.

The contribution of this thesis is therefore a problem-specific comparison rather than a claim of universal algorithm superiority. The results provide evidence for selecting optimization methods for the suction-layout and rigid-jaw-contour formulations considered here.

3 Methodology

This chapter presents the methodological foundations of the gripper optimization study. It describes the research design, problem formulation, geometric parameterizations, objective criteria, selected optimization frameworks and algorithms, workpiece selection, evaluation architecture, and performance indicators. Implementation details and the specific experimental procedure are presented separately in the approach chapter.

3.1 Research Design

3.1.1 Research Paradigm

In this thesis, gripper configurations are represented by a parameterized mathematical model, evaluated by means of computational geometry, and improved through optimization runs that access this evaluation only as a black box. This procedure follows from the nature of the problem itself: the quality of a gripper configuration results from the geometric interaction between the gripper and the workpiece, which depends on the specific shapes involved and is therefore not available in closed analytical form. Consequently, no gradients or structural properties of the objective functions can be exploited, and each candidate configuration has to be assessed by explicit geometric evaluation.

3.1.2 Study Structure

The suction-cup and parallel-jaw grippers are treated as separate optimization studies. Each optimization run considers exactly one workpiece. Different workpieces therefore define separate problem instances rather than simultaneous objectives within the same optimization problem.

For the suction-cup gripper, the positions of three suction cups are optimized for the selected workpiece. The optimization considers the spatial and orientational coverage of the resulting valid grasps, the number of valid grasps and the area of the triangle formed by the three suction cups.

For the parallel-jaw gripper, a spline-based jaw contour is optimized for a predefined set of grasps on the selected workpiece. The contour is evaluated according to how well it conforms to the workpiece geometry associated with the complete predefined grasp set while avoiding geometrically infeasible contact conditions.

Because the two gripper studies use different geometric representations and evaluation criteria, their optimization results are analysed separately and are not combined into a common Pareto front.

3.1.3 Experimental Setup

Each combination of gripper type and workpiece defines a separate optimization instance. The selected algorithms are applied independently to each instance under comparable conditions. Multiple independent repetitions allow the variability of the stochastic optimization methods to be examined. Within an instance, the algorithms use the same decision-variable bounds, objective definitions, feasibility conditions, and evaluation budget.

3.1.4 Scope and Limitations

The conclusions of the study are limited to the selected gripper parameterizations, workpieces, objective criteria, and evaluation budgets. Although repeated computational runs allow performance variability to be examined, the results do not establish that one algorithm is universally superior across all gripper-design problems.

The evaluation is entirely computational. Manufacturing effects, material properties, contact forces, suction behavior, and experimental validation on physical grippers are outside the scope of the thesis. Consequently, the optimized configurations are evaluated as geometric designs rather than as fully validated physical grippers.

3.2 Problem Formulation

3.2.1 Black-Box Optimization Framework

The gripper-design problems are formulated as constrained black-box optimization problems. Let $\mathbf{q} \in \mathcal{Q}$ denote a candidate gripper configuration and let d denote the number of decision variables. The general vector-valued optimization problem is written as

$$\min_{\mathbf{q}\in\mathcal{Q}} \mathbf{f}(\mathbf{q}) = (f_1(\mathbf{q}), f_2(\mathbf{q}), \ldots, f_m(\mathbf{q}))\,, \tag{3.1}$$

subject to the inequality constraints

$$g_j(\mathbf{q}) \leq 0, \qquad j = 1, \ldots, n_c, \tag{3.2}$$

and the variable bounds

$$q_i^{\mathrm{L}} \leq q_i \leq q_i^{\mathrm{U}}, \qquad i = 1, \ldots, d. \tag{3.3}$$

Objectives that are naturally maximized are transformed to maintain a common minimization convention. For problems with conflicting criteria, a multi-objective formulation preserves alternative compromise solutions, while weighted scalarization collapses the problem to a single solution defined by chosen weights[81].

The number of objectives is determined by the performance criteria of the corresponding gripper problem. It is not determined by the number of workpieces included in the overall study. Each workpiece is optimized independently.

3.2.2 Parameterization of Gripper Geometry

A central methodological decision is the representation of gripper geometry as a finite-dimensional decision vector. The representation must provide sufficient geometric flexibility while remaining compact, interpretable, and suitable for black-box optimization [82, 5, 83].

Suction-Cup Gripper Representation

For the suction-cup gripper, representing the planar Cartesian coordinates of three cups independently would require six decision variables. A reduced polar parameterization is therefore selected. The position of cup i is expressed as

$$\mathbf{p}_i = \begin{pmatrix} r\cos\alpha_i \\ r\sin\alpha_i \end{pmatrix}, \qquad i = 1,2,3, \tag{3.4}$$

which gives the decision vector

$$\mathbf{q}_{\mathrm{SC}} = (r,\alpha_1,\alpha_2,\alpha_3)\,. \tag{3.5}$$

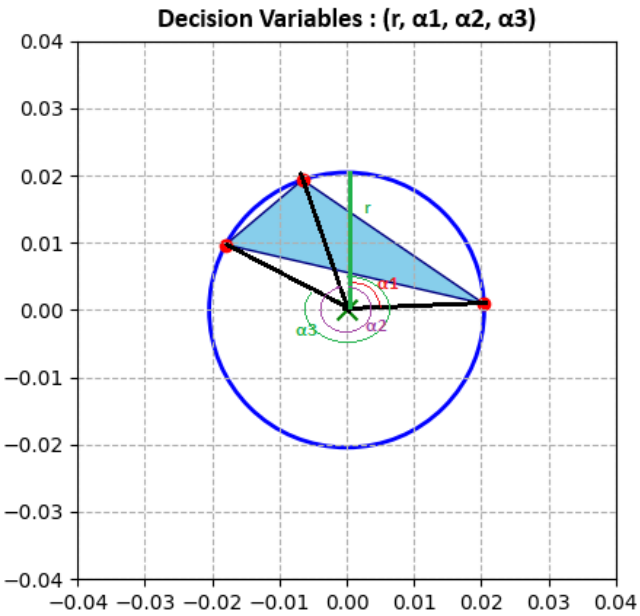


Figure 3.1: polar parameterization of Suction cup gripper.

Figure 3.1 shows the polar coordinates of a potential suction cup gripper. The common radius r determines the distance of the cups from the gripper origin, while the angles α_i

determine their arrangement. This representation reduces the dimensionality of the search space and restricts the cup centers to a common circle. Separation constraints are required to prevent overlapping or degenerate cup arrangements.

Parallel-Jaw Gripper and Contour Representation

The parallel-jaw gripper applies gripping forces through two opposing jaw surfaces. The geometry of these surfaces determines how closely the gripper conforms to the workpiece at the predefined grasps. The jaw contour therefore requires a representation that is compact enough for optimization while remaining expressive enough to describe smooth and physically realizable shapes [9].

A cubic interpolating spline is used to represent the jaw contour. Such a spline is a piecewise polynomial curve of degree at most three that passes through a prescribed set of interpolation points. Adjacent polynomial segments are connected so that the curve and its first and second derivatives remain continuous. This produces a smooth contour, while appropriate endpoint conditions determine the behavior of the spline boundaries.

Spline parameterization reduces the continuous shape-design problem to a finite-dimensional optimization problem. The endpoint interpolation points are fixed to preserve the required spatial extent of the jaw, while three interior interpolation points are variable and define the contour decision variables. Increasing the number of variable points would allow more complex contours but would also increase the dimensionality of the optimization problem [84, 85].

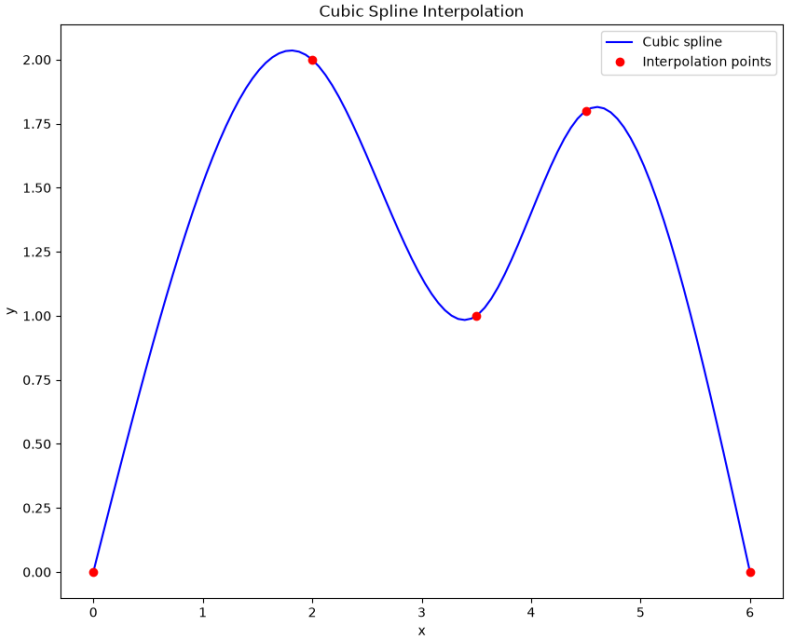


Figure 3.2: Cubic spline interpolation through prescribed interpolation points.

Figure 3.2 illustrates a cubic interpolating spline passing through its prescribed data points. In contrast to a B-spline control-point representation, the interpolation points lie directly on the resulting curve. Consequently, the three variable interior points provide intuitive geometric parameters for modifying the contour, while the fixed endpoints preserve its prescribed spatial extent.

3.2.3 Selection of Objective Functions

The objective functions define the properties used to distinguish effective gripper configurations. They must represent meaningful geometric performance criteria while remaining computable through the available evaluation models [5, 86].

For the multi-suction cup gripper, three objectives are used.

Grasp diversity The first objective is the diversity of the valid grasps in position and orientation. A layout that can grasp the workpiece in only one region is of limited use when the part arrives in varying poses or when parts of its surface are blocked by fixtures or neighbouring objects. A diversity measure captures this operational flexibility [87, 88].

Number of distinct grasps Diversity alone does not show how many alternatives exist, because two well-separated grasps can already produce a large spread. The number of distinct valid grasps is therefore the second objective, as a measure of robustness: the more distinct grasps a layout offers, the more likely a usable one remains when others are excluded by the scene or by reachability. Only distinct grasps are counted, so that near-identical grasps cannot inflate the score.

Support area The third objective is the area of the triangle spanned by the three cups. A wider support polygon gives a longer lever arm against tilting moments, so the area serves as a stability proxy [89]. It is a purely geometric measure. Suction force, sealing quality, material properties and the centre of mass are not modelled. The area was also chosen because it conflicts with the first two objectives. A wide layout needs larger surface regions on which all three cups can seal, which reduces both the number and the spread of feasible grasps. This conflict is the central trade-off of the suction-cup design and is why a multi-objective formulation is needed. All three objectives are normalized so that they remain comparable across workpieces and within the hypervolume indicator.

For the parallel-jaw gripper, the objective criterion evaluates the conformity of a candidate spline contour to a predefined set of grasps on one workpiece. The contour should provide close geometric correspondence with the workpiece across the complete grasp set rather than being specialized for only one grasp. A voxel-based convolution method developed at Fraunhofer IPA is used to assess finger-to-workpiece proximity and contact quality while accounting for geometrically infeasible or colliding configurations [82, 90].

3.3 Optimization Library Selection

The evolutionary optimization algorithms are selected from the `pymoo` framework. `pymoo` provides a unified Python interface for multi-objective evolutionary optimization, constraint

handling, performance evaluation, and visualization. Using algorithms from a common framework reduces implementation differences and improves the reproducibility of the comparison. The framework also provides established multi-objective performance indicators, including hypervolume and inverted generational distance plus. [64]

Bayesian multi-objective optimization is represented using the `Ax` platform and its underlying `BoTorch` functionality. Bayesian optimization constructs probabilistic surrogate models of the black-box objective functions and uses an acquisition function to select promising candidate evaluations. `Ax` provides the experiment-management interface, while `BoTorch` provides the probabilistic modeling and acquisition-function capabilities. [3, 91]

The use of established open-source frameworks avoids the need to implement the optimization methods, surrogate models, and acquisition functions from scratch. It also supports a more reproducible comparison between evolutionary and Bayesian optimization strategies.

3.4 Algorithm Selection

For the experimental comparison, NSGA-II, NSGA-III, MOEA/D, SMS-EMOA, CMOPSO, and multi-objective Bayesian optimization are selected.

The selection was made to compare different approaches to multi-objective search under the same gripper-design problems and evaluation budget. Including algorithms with different search principles reduces the extent to which the comparison is limited to variations of a single optimization strategy. It also allows the experiments to determine whether the relative performance depends on the selected gripper type or workpiece.

- **NSGA-II** is selected as the baseline algorithm for the population-based comparison.
- **NSGA-III** is included to compare the NSGA-II baseline with an alternative that organizes the search using predefined reference directions.
- **MOEA/D** is selected to include a decomposition-based approach and to examine how this formulation performs on the objective trade-offs of the gripper-design problems.
- **SMS-EMOA** is included as an indicator-based approach. Its selection also permits investigation of an algorithm whose search is directly associated with hypervolume, one of the performance indicators used in this thesis.

- **CMOPSO** is selected to include a swarm-based search method and thereby extend the population-based comparison beyond evolutionary algorithms.
- **Multi-objective Bayesian optimization** is included as a model-based alternative to the population-based methods. This allows its performance under a limited evaluation budget to be compared with that of the other selected algorithms.

Together, these algorithms form a deliberately varied comparison set containing dominance-based, reference-direction-based, decomposition-based, indicator-based, swarm-based, and model-based approaches. The experiments use this selection to determine which approaches provide suitable convergence, trade-off representation, and consistency for the suction-cup and parallel-jaw gripper-design problems.

3.5 Workpiece Selection

The workpieces are selected to provide geometrically diverse problem instances rather than to optimize the grippers for one particular object. Geometric diversity is important because algorithm performance can depend on the structure of the feasible design space and the resulting optimization landscape [92, 93]. Four workpieces are used, covering planar, cylindrical, elongated, and free-form geometries.

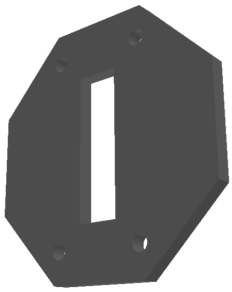


Figure 3.3: IPA Intermediate Plate workpiece

The **IPAIntermediatePlate** shown in figure 3.3 is an industrial component with regular geometry, large planar regions, and straight edges. It covers the class of structured, prismatic workpieces commonly found in manufacturing and material-handling applications.

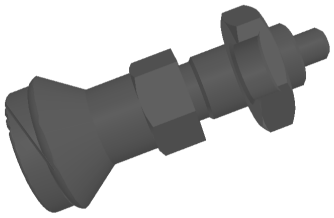


Figure 3.4: Knob workpiece

The **Knob** shown in figure 3.4 is a predominantly cylindrical, rotationally symmetric part with a stepped circumferential profile.



Figure 3.5: Connecting rod workpiece

The **ConnectingRod** shown in figure 3.5 is a flat, strongly elongated component with a high length-to-width ratio, consisting of two circular ends connected by a narrow shaft. It

represents the class of slender, asymmetric workpieces whose principal dimensions differ by a large factor.



Figure 3.6: Stanford Bunny workpiece

The **Stanford Bunny** shown in figure 3.6 is a free-form object with curved surfaces, irregular features, and considerable local geometric variation. It covers the class of organic, non-symmetric geometries without planar or axisymmetric structure.

Each workpiece defines an independent optimization instance. The workpieces are not combined as objectives within a single optimization run. Their purpose is to evaluate whether the relative behavior of the selected optimization methods changes across different workpiece geometries.

3.6 Evaluation Architecture

Candidate gripper configurations are evaluated through a client–server architecture. The optimization process acts as the client and submits a decision vector $\mathbf{q}$ to the evaluation server. The server evaluates the corresponding gripper geometry and returns the objective vector $\mathbf{f}(\mathbf{q})$.

This separation establishes a strict black-box interface. The optimization algorithms operate only on decision vectors and returned objective values and do not access the internal

geometric evaluation procedure. Consequently, algorithms that support the same interface can be exchanged without modifying the geometric evaluation pipelines. The architecture therefore supports modularity, reproducibility, and future extension.

3.7 Comparative Experimental Design

A controlled comparative design is adopted to ensure that observed differences are attributable primarily to the optimization methods. For a given gripper–workpiece instance, all algorithms are evaluated using the same geometric parameterization, objective functions, variable bounds, feasibility conditions, and evaluation budget.

Because the selected optimization methods contain stochastic components, independent repetitions are used to examine performance variability and robustness [94]. Comparisons are performed within the same problem instance. Results from different workpieces or gripper types are not combined into one optimization problem or one Pareto front.

3.8 Performance Indicators

Algorithm performance is evaluated using inverted generational distance plus and the hypervolume attained during a run.

IGD+ is selected to assess how well an optimization run approximates the best trade-offs identified for a given gripper–workpiece problem. It is useful in this thesis because it considers both proximity to the reference front and the representation of different regions of that front. This enables the algorithms to be compared based on the overall quality of their approximation sets rather than on individual objective values alone.

Hypervolume is selected as an aggregate measure of approximation-set quality. It provides a single value that reflects both convergence and the extent of the represented objective-space trade-offs. Considering the final value attained during a run is particularly relevant for the investigated black-box optimization problems because it identifies the best approximation quality reached within the available optimization budget, even when the quality of the population varies during a run.

The two performance indicators are selected because they rely on different reference information: IGD+ measures proximity to and coverage of a reference front assembled from all runs, whereas the hypervolume requires only a reference point and measures the objective-space region dominated by the non-dominated solutions found during a run. Their values across independent repetitions are used to characterize both optimization performance and its variability.

4 Implementation

This chapter presents the optimization frameworks developed for two gripper types: a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration, and a two-finger gripper with a spline-based gripping contour. Both frameworks support evolutionary multi-objective optimization and Bayesian multi-objective optimization. The evolutionary algorithms are implemented using the `pymoo` library [64], whereas Bayesian optimization is implemented using the `Ax` platform and its underlying `BoTorch` modeling framework [3, 91].

For both gripper types and both optimization approaches, the evaluation of candidate gripper designs is delegated to an external service through an API. This separation provides a modular architecture in which the optimization strategy and the domain-specific grasp evaluation remain decoupled. Consequently, the same decision-variable representation, geometry-generation pipeline, grasp-generation procedure, and objective functions can be used by both evolutionary and Bayesian optimization methods, allowing their performance to be compared under a common evaluation budget and a common evaluation pipeline.

4.1 Implementation of the Common Gripper Optimization Framework

This section describes the parts of the implementation that are shared by both problems: the configuration, the supported optimization methods, the initialization, the optimization loops, the termination criteria, the performance monitoring, the optional hyperparameter optimization, the objective-direction conventions, and the output. The problem-specific parameterizations, geometry pipelines, and objective definitions are described in Section 4.2 for the multi-suction cup gripper and in Section 4.3 for the two-finger gripper. An overview of the two problems is given in Table 4.1.

Table 4.1: Overview of the two gripper optimization problems.

	Multi-suction cup gripper	**Two-finger gripper**
Decision vector	$\mathbf{q} = [r,\alpha_1,\alpha_2,\alpha_3]$	$\mathbf{q} = [\ell_{\text{gn}},h_1,h_2,h_3]^{\text{T}}$
Dimension d	4	4
Objectives	$m = 3$: diversity D, normalised grasp count $\hat{n}_{\text{g}}$, normalised triangle area $\hat{A}$	$m = n_{\text{or}} = 10$: one proximity KPI k_i per predefined workpiece orientation
Evaluation core	Suction grasp generation, seal-formation and collision filtering, clustering	CadQuery geometry generation, voxelization, FFT-based collision and proximity analysis

4.1.1 Optimization Framework Configuration

Each gripper problem is solved using either an evolutionary multi-objective optimization algorithm or Bayesian multi-objective optimization. Both approaches operate on the same decision vector, variable bounds, objective functions, and external evaluation service.

The common optimization configuration contains:

- Number of objectives m
- Decision-variable definitions and bounds
- Optimization direction (minimization or maximization)
- Choice of optimization method
- Evaluation endpoint URL

For evolutionary optimization, the additional configuration parameters include:

- Population size N_{pop}
- Maximum number of generations G_{max}
- Choice of evolutionary algorithm
- Algorithm-specific parameters, such as crossover and mutation settings

For Bayesian optimization, the additional configuration parameters include:

- Maximum number of candidate-generation batches $B_{\max}$, which is read from the same configuration field as the number of generations of the evolutionary algorithms
- Maximum number of candidates requested per batch b

In the current Bayesian implementation, the configuration value named `generations` represents the number of candidate-generation batches rather than evolutionary generations. A batch therefore represents a candidate-generation step rather than an evolutionary generation. The maximum requested number of Bayesian evaluations is

$$E_{\max}^{\mathrm{BO}} = B_{\max}\, b. \tag{4.1}$$

The actual number of evaluations may be smaller if the generation strategy returns fewer than b candidates in a particular batch.

Supported Optimization Methods

The framework supports the evolutionary and Bayesian multi-objective optimization methods listed in Table 4.2.

Table 4.2: Supported multi-objective optimization methods.

Family	**Identifier**	**Method**
Evolutionary	NSGA2	Non-dominated Sorting Genetic Algorithm II
Evolutionary	NSGA3	Non-dominated Sorting Genetic Algorithm III
Evolutionary	MOEAD	Multi-Objective Evolutionary Algorithm based on Decomposition
Evolutionary	CMOPSO	Competitive Multi-Objective Particle Swarm Optimization
Evolutionary	SMSEMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm
Bayesian	AX-MBM	Ax modular BoTorch model for multi-objective Bayesian optimization

4.1.2 Initialization

The initialization procedure depends on the selected optimization approach.

Evolutionary Initialization

For evolutionary optimization, an initial population of N_{pop} candidate solutions is generated by sampling the decision variables within their bounds:

$$\begin{gathered} \mathcal{U}^{(0)} = \left\{ \mathbf{q}_1^{(0)}, \mathbf{q}_2^{(0)}, \ldots, \mathbf{q}_{N_{\text{pop}}}^{(0)} \right\}, \\ \mathcal{Q} \subseteq \mathbb{R}^d, \quad d \in \mathbb{N}, \quad \mathbf{q}_k^{(0)} \in \mathcal{Q}, \quad k \in \{1, \ldots, N_{\text{pop}}\}, \quad N_{\text{pop}} \in \mathbb{N}, \\ \mathcal{U}^{(0)} \subset \mathcal{Q}, \quad |\mathcal{U}^{(0)}| = N_{\text{pop}} \end{gathered} \tag{4.2}$$

The complete population is evaluated through the external API before the selected evolutionary algorithm performs its first selection and variation step. For the multi-suction cup gripper, each candidate decision vector $\mathbf{q}_k$ is transformed into its corresponding Cartesian gripper configuration $\mathbf{P}_k$ before objective evaluation, as described in Section 4.2.3.

Bayesian Initialization

Bayesian optimization does not maintain an evolutionary population. Instead, it stores a dataset of completed trials:

$$\begin{gathered} \mathcal{D}_t = \{(\mathbf{q}_k, \mathbf{f}(\mathbf{q}_k))\}, \qquad k = 1,2,\ldots,t, \\ \mathbf{q}_k \in \mathcal{Q} \subset \mathbb{R}^d, \quad \mathbf{f} : \mathcal{Q} \to \mathcal{Y} \subseteq \mathbb{R}^m, \quad m \in \mathbb{N}, m \geq 2, \quad t \in \mathbb{N}_0, \\ \mathcal{D}_t \subset \mathcal{Q} \times \mathcal{Y} \text{ (multiset of } t \text{ trials)}, \quad \mathcal{D}_0 = \emptyset, \quad \mathcal{D}_t \subseteq \mathcal{D}_{t+1} \end{gathered} \tag{4.3}$$

where $\mathbf{q}_k$ is an evaluated gripper parameterization and $\mathbf{f}(\mathbf{q}_k)$ is its observed objective vector as returned by the evaluation service.

The Ax generation strategy first evaluates the center of the bounded search space and then generates an initial set of candidates using Sobol quasi-random sampling. Sobol sampling provides broad and approximately space-filling coverage of the four-dimensional decision space before a surrogate model is fitted. Once the initialization requirements of the generation strategy have been satisfied, Ax transitions to the model-based modular BoTorch model (MBM) stage. From this point onward, new gripper designs are selected using surrogate predictions and predictive uncertainty.

4.1.3 Objective-Direction Conventions

For both gripper problems, all objectives are maximised in their natural form, so that the natural optimization problem is

$$\max_{\mathbf{q}\in\mathcal{Q}} \mathbf{f}(\mathbf{q}), \tag{4.4}$$

where $\mathcal{Q}$ is the bounded decision space introduced in Section 3.2.1.

The objective-direction convention is handled differently by the two optimization frameworks. Since `pymoo` assumes minimisation, the evolutionary algorithms receive the negated objective vector

$$\mathbf{f}_{\mathrm{EA}}(\mathbf{q}) = -\,\mathbf{f}(\mathbf{q}). \tag{4.5}$$

In the Bayesian implementation, the values returned by the external evaluation service are left unchanged, since the maximisation direction is declared through the `Ax` optimization configuration:

$$\mathbf{f}_{\mathrm{BO}}(\mathbf{q}) = \mathbf{f}(\mathbf{q}). \tag{4.6}$$

Consequently, the observations are not negated before they are attached to the corresponding Ax trial, which preserves the original physical meaning of the objective values in the returned Pareto set. For performance comparisons, the results of both optimization frameworks are converted to the common natural maximisation convention $\mathbf{f}(\mathbf{q})$, so that all reported objective values correspond to the original direction and magnitude of the underlying quality measures.

4.1.4 Optimization Loop

The external evaluation pipeline is shared by the evolutionary and Bayesian optimization approaches. The methods differ primarily in how they use the completed evaluations to generate subsequent candidates, while the external candidate-evaluation procedure remains identical.

For every candidate, the common evaluation steps are:

1. Send the candidate decision vector to the external evaluation API.
2. Map the decision variables to the complete problem-specific gripper configuration.
3. Evaluate the candidate using the problem-specific grasp-evaluation pipeline.
4. Return the objective vector to the selected optimization framework.

The problem-specific instantiations of these steps are given in Section 4.2.3 and Section 4.3.2.

Evolutionary Optimization Loop

In every evolutionary generation, the selected algorithm generates a population of candidate decision vectors. The candidates are evaluated through the common evaluation pipeline, and the resulting objective values are negated before being supplied to `pymoo`. The population update can be expressed abstractly as

$$\begin{gathered} \mathcal{U}^{(G+1)} = \mathcal{A}\left(\mathcal{U}^{(G)}, \mathbf{F}_{\mathrm{EA}}^{(G)}\right), \qquad G = 1,2,\ldots,G_{\max}, \\ \mathcal{U}^{(G)} \subset \mathcal{Q} \subset \mathbb{R}^d, \quad |\mathcal{U}^{(G)}| \leq N_{\mathrm{pop}}, \quad \mathbf{F}_{\mathrm{EA}}^{(G)} \in \mathbb{R}^{n^{(G)} \times m}, \quad n^{(G)} \in \mathbb{N}, \\ \mathcal{A} : \mathcal{Q}^{N_{\mathrm{pop}}} \times \mathbb{R}^{n^{(G)} \times m} \times \Omega \to \mathcal{Q}^{N_{\mathrm{pop}}}, \quad G \in \mathbb{N}, \quad G_{\max} \in \mathbb{N} \end{gathered} \tag{4.7}$$

where $\mathcal{A}(\cdot,\cdot)$ denotes the selected evolutionary algorithm, which maps the current population and the minimisation-form objective values onto the next population, and $\mathbf{F}_{\mathrm{EA}}^{(G)}$ contains the minimisation-form objective vectors evaluated in generation G.

Depending on the selected algorithm, the new population is generated through operations such as selection, crossover, mutation, decomposition, particle movement, or survival based on hypervolume contribution as explained in Section 2.1.3. Every candidate in the resulting population is evaluated through the common API-based evaluation pipeline.

Bayesian Optimization Loop

The Bayesian optimization procedure maintains the observation dataset $\mathcal{D}_t$. During the initialization phase, candidates are generated from the center of the search space and by Sobol sampling. During the model-based stage, Ax and BoTorch use the completed trial dataset to fit probabilistic surrogate models:

$$\begin{gathered}\widehat{\mathbf{f}}_t = \mathcal{M}(\mathcal{D}_t),\\ \mathcal{M} : \bigcup_{\tau \geq t_{\text{init}}} (\mathcal{Q} \times \mathcal{Y})^{\tau} \to \{\, \widehat{\mathbf{f}} : \mathcal{Q} \to \Delta(\mathbb{R}^m) \,\}, \qquad t \geq t_{\text{init}},\\ \widehat{\mathbf{f}}_t(\mathbf{q}) \sim \mathcal{N}(\boldsymbol{\mu}_t(\mathbf{q}), \boldsymbol{\Sigma}_t(\mathbf{q}))\,, \quad \boldsymbol{\mu}_t : \mathcal{Q} \to \mathbb{R}^m, \quad \boldsymbol{\Sigma}_t : \mathcal{Q} \to \mathbb{S}_+^m\end{gathered} \tag{4.8}$$

where $\mathcal{M}(\cdot)$ denotes the surrogate-modeling procedure used by Ax and BoTorch.

The surrogate models approximate the relationship between the gripper parameters and the objective values. In addition to predicted objective values, they estimate predictive uncertainty in regions containing few or no observations. An acquisition function $\psi(\cdot \mid \mathcal{D}_t)$ uses this information to balance:

- Exploitation of gripper designs expected to improve the current Pareto frontier
- Exploration of uncertain regions of the bounded decision space

For a requested batch size b, the next candidate batch is generated as

$$\begin{gathered}\mathcal{B}_{t+1} = \underset{\substack{\mathcal{B} \subset \mathcal{Q} \\ |\mathcal{B}| \leq b}}{\arg\max}\, \psi(\mathcal{B} \mid \mathcal{D}_t),\\ \psi(\,\cdot \mid \mathcal{D}_t) : \{\mathcal{B} \subset \mathcal{Q} : |\mathcal{B}| \leq b\} \to \mathbb{R}, \quad b \in \mathbb{N},\\ \mathcal{B}_{t+1} \subset \mathcal{Q} \subset \mathbb{R}^d, \quad 0 \leq |\mathcal{B}_{t+1}| \leq b \text{ (the strategy may return fewer than } b \text{ candidates)}\end{gathered} \tag{4.9}$$

Each candidate $\mathbf{q} \in \mathcal{B}_{t+1}$ is evaluated through the external evaluation service, and the resulting observations are supplied to Ax as completed trials. After all candidates of the generated batch have been evaluated, the dataset is updated as

$$\mathcal{D}_{t+|\mathcal{B}_{t+1}|} = \mathcal{D}_t \cup \{(\mathbf{q}, \mathbf{f}(\mathbf{q})) \mid \mathbf{q} \in \mathcal{B}_{t+1}\}\,. \tag{4.10}$$

The surrogate models are subsequently updated using the expanded dataset, and another candidate batch is generated. This process is repeated until the specified evaluation budget is exhausted.

Although multiple candidates can be requested in one Bayesian batch, the current implementation submits the candidates of that batch one after another to the external evaluation service. The implementation therefore performs *sequential batch evaluation* rather than parallel batch evaluation.

4.1.5 Termination Criteria

The termination criteria depend on the selected optimization approach and are summarised in Table 4.3.

Table 4.3: Termination criteria for evolutionary and Bayesian optimization.

Approach	**Criterion**	**Symbol**	**Description**
Evolutionary	Design-space tolerance	q_{tol}	Change in decision variables below threshold
Evolutionary	Objective-space tolerance	f_{tol}	Relative objective improvement below threshold
Evolutionary	Maximum generations	$G_{\max}$	Maximum number of generations reached
Evolutionary	Maximum evaluations	$E_{\max}$	Maximum number of function evaluations reached
Evolutionary	Hypervolume stagnation	$\Delta\text{HV}_{\text{tol}}$	Hypervolume improvement over the monitoring window below threshold
Bayesian	Maximum batches	$B_{\max}$	Maximum number of candidate-generation batches reached
Bayesian	Maximum evaluations	$E_{\max}^{\text{BO}}$	Bayesian evaluation budget exhausted

For evolutionary optimization, the design-space and objective-space tolerances are evaluated over a sliding window of n_{win} generations.

In the current Bayesian implementation, termination is budget-based. The optimization performs at most $B_{\max}$ candidate-generation batches, with up to b candidates requested per

batch. Therefore,

$$E_{\max}^{\mathrm{BO}} = B_{\max}\, b. \tag{4.11}$$

No design-space or objective-space convergence tolerance is currently applied to the Bayesian optimization loop.

4.1.6 Performance Monitoring

The hypervolume indicator is used to monitor both evolutionary and Bayesian optimization. Before calculating the hypervolume, all objective vectors are represented using the common natural maximisation convention $\mathbf{f}(\mathbf{q})$ introduced in Section 4.1.3.

For evolutionary optimization, a callback records the optimization state after each generation, so that the hypervolume indicator is tracked per generation:

$$\begin{gathered}
\mathrm{HV}^{(G)} = \mathrm{HV}\Big(\mathcal{F}^{(G)}, \mathbf{z}^{\mathrm{ref}}\Big), \\
\mathcal{F}^{(G)} = \Big\{\mathbf{f}(\mathbf{q}) \;\Big|\; \mathbf{q} \in \mathcal{U}^{(G)}\Big\} \subset \mathcal{Y} \subseteq \mathbb{R}^m, \quad |\mathcal{F}^{(G)}| \le N_{\mathrm{pop}}, \\
\mathbf{z}^{\mathrm{ref}} \in \mathbb{R}^m \text{ with } z_i^{\mathrm{ref}} \le \inf_{\mathbf{q}\in\mathcal{Q}} f_i(\mathbf{q}),\ i = 1, \ldots, m, \\
\mathrm{HV} : \big\{\mathcal{F} \subset \mathbb{R}^m : |\mathcal{F}| < \infty\big\} \times \mathbb{R}^m \to \mathbb{R}_{\ge 0}, \quad \mathrm{HV}(\emptyset, \mathbf{z}^{\mathrm{ref}}) = 0
\end{gathered} \tag{4.12}$$

where $\mathbf{z}^{\mathrm{ref}} \in \mathbb{R}^m$ is the reference point and $\mathcal{F}^{(G)}$ denotes the set of objective vectors associated with the population $\mathcal{U}^{(G)}$.

Equivalently, when only the non-dominated objective vectors available after generation G are retained, denoted $\mathcal{F}_{\mathrm{EA}}^{(G)}$, the corresponding hypervolume is

$$\begin{gathered}
\mathrm{HV}_{\mathrm{EA}}^{(G)} = \mathrm{HV}\left(\mathcal{F}_{\mathrm{EA}}^{(G)}, \mathbf{z}^{\mathrm{ref}}\right), \\
\mathcal{F}_{\mathrm{EA}}^{(G)} \subseteq \mathcal{F}^{(G)} \subset \mathbb{R}^m \text{ (non-dominated subset)}, \quad 1 \le |\mathcal{F}_{\mathrm{EA}}^{(G)}| \le N_{\mathrm{pop}}, \\
\mathrm{HV}_{\mathrm{EA}}^{(G)} \in \mathbb{R}_{\ge 0}; \quad \mathrm{HV}_{\mathrm{EA}}^{(G+1)} \ge \mathrm{HV}_{\mathrm{EA}}^{(G)} \text{ only if } \mathcal{F}_{\mathrm{EA}}^{(G)} \text{ is a cumulative archive}
\end{gathered} \tag{4.13}$$

For Bayesian optimization, the hypervolume can be calculated after every completed trial or after every completed candidate batch. Let $\mathcal{F}_{\mathrm{BO}}^{(t)}$ denote the non-dominated objective vectors

obtained from the first t completed Bayesian trials. The corresponding hypervolume is

$$\begin{gathered} \mathrm{HV}_{\mathrm{BO}}^{(t)} = \mathrm{HV}\left(\mathcal{F}_{\mathrm{BO}}^{(t)}, \mathbf{z}^{\mathrm{ref}}\right), \\ \mathcal{F}_{\mathrm{BO}}^{(t)} \subset \mathbb{R}^m, \quad |\mathcal{F}_{\mathrm{BO}}^{(t)}| \leq t, \quad t \in \mathbb{N}_0, \quad \mathrm{HV}_{\mathrm{BO}}^{(0)} = 0, \\ \mathrm{HV}_{\mathrm{BO}}^{(t+1)} \geq \mathrm{HV}_{\mathrm{BO}}^{(t)} \text{ (cumulative archive by construction)} \end{gathered} \tag{4.14}$$

Under the maximisation convention, the fixed reference point $\mathbf{z}^{\mathrm{ref}}$ must be worse than the objective values of interest. The same reference point and objective scaling must be used for all compared methods. An increasing hypervolume generally indicates progress toward a better approximation of the Pareto front, improved coverage of the objective space, or both. Monotonicity is guaranteed only if the indicator is computed from a cumulative non-dominated archive; the hypervolume of the current population may fluctuate between generations.

Evolutionary generations and Bayesian batches can contain different numbers of candidate evaluations. Therefore, the primary comparison is performed using the hypervolume as a function of the cumulative number of objective evaluations,

$$\begin{gathered} \mathrm{HV} = \mathrm{HV}(E), \\ \mathrm{HV}(\cdot) : \{0,1,\ldots,E_{\max}^{\bullet}\} \to \mathbb{R}_{\geq 0}, \qquad \bullet \in \{\mathrm{EA},\mathrm{BO}\}, \\ E \in \mathbb{N}_0, \quad \mathrm{HV}(0) = 0 \end{gathered} \tag{4.15}$$

where E is the number of completed external API evaluations.

4.1.7 Optional Hyperparameter Optimization

For the evolutionary algorithms, the framework optionally supports automated tuning of algorithm hyperparameters, such as crossover probability, mutation rate, mutation eta, crossover eta, selection eta, etc., using a meta-optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The negative mean hypervolume across the runs is used as the

meta-objective:

$$\begin{gathered}\min_{\boldsymbol{\theta}\in\Theta}\left\{-\frac{1}{|\mathcal{Z}|}\sum_{\sigma\in\mathcal{Z}}\mathrm{HV}\Big(\mathcal{F}_\sigma(\boldsymbol{\theta}),\mathbf{z}^{\mathrm{ref}}\Big)\right\},\\ \Theta=\Theta_{\mathrm{cont}}\times\Theta_{\mathrm{int}}\times\Theta_{\mathrm{cat}}\ \text{(mixed hyperparameter space)},\quad \mathcal{Z}\subset\mathbb{N},\ |\mathcal{Z}|\in\mathbb{N},\\ \mathcal{F}_\sigma(\boldsymbol{\theta})\subset\mathbb{R}^m\ \text{finite},\quad \text{objective value}\in[-\mathrm{HV}_{\max},0]\end{gathered}\tag{4.16}$$

The best hyperparameter configuration is subsequently applied to the main evolutionary optimization run.

The current Bayesian implementation uses the generation strategy, surrogate configuration, and acquisition-function configuration selected by Ax. A separate meta-optimization of Bayesian model hyperparameters is therefore not currently performed. The user specifies the candidate batch size and the optimization budget, while Ax manages the transition from initialization to model-based optimization.

4.1.8 Output

Because all objectives are maximised in their natural form, the optimization produces the set of Pareto-optimal solutions

$$\mathcal{P}^*=\left\{\mathbf{q}\in\mathcal{Q}\mid \nexists\,\mathbf{q}'\in\mathcal{Q}\text{ such that }\mathbf{f}(\mathbf{q}')\geq\mathbf{f}(\mathbf{q})\text{ component-wise and }\mathbf{f}(\mathbf{q}')\neq\mathbf{f}(\mathbf{q})\right\}.\tag{4.17}$$

Equivalently, $\mathbf{q}'\succ\mathbf{q}$ denotes that $\mathbf{q}'$ dominates $\mathbf{q}$, i.e.

$$\mathcal{P}^*=\left\{\mathbf{q}\in\mathcal{Q}\mid \nexists\,\mathbf{q}'\in\mathcal{Q}:\mathbf{q}'\succ\mathbf{q}\right\},\tag{4.18}$$

with

$$f_i(\mathbf{q}')\geq f_i(\mathbf{q})\quad\forall\,i=1,\ldots,m\tag{4.19}$$

and

$$f_j(\mathbf{q}')>f_j(\mathbf{q})\quad\text{for at least one objective } j.\tag{4.20}$$

The final output of an optimization run includes:

- The optimization configuration used to carry out the optimization from Section 4.1.1
- The optimization algorithm hyperparameters
- The Pareto-optimal set $\mathcal{P}^*$ in the decision space
- The corresponding objective values $\mathbf{f}(\mathbf{q})$
- The hypervolume progression
- Total optimization wall-clock time

The problem-specific contents and interpretations of the returned decision vectors are given in Section 4.2.5 and Section 4.3.4.

4.2 Implementation of Multi-Suction Cup Gripper Optimization

This section describes the problem-specific implementation for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration. The common framework components are described in Section 4.1.

4.2.1 Reduced Parameterization of Multi-Suction Cup Gripper

The gripper consists of three suction cups whose positions must ultimately be represented in Cartesian space in order to perform grasp generation, collision checking, and objective computation. However, the Cartesian suction cup coordinates are not used directly as optimization variables. Instead, the layout is parameterized in the reduced form introduced in Section 3.2.2 in order to constrain the search space and encode geometric structure.

All three suction cups are constrained to lie on a common circle in the gripper plane. Their configuration is therefore defined by one shared radius r and three angular parameters α_1,α_2,α_3:

$$\mathbf{q}_{\mathrm{SC}} = [r,\alpha_1,\alpha_2,\alpha_3]. \tag{4.21}$$

For brevity, $\mathbf{q}$ denotes the suction-cup decision vector $\mathbf{q}_{\mathrm{SC}}$ throughout this section.

The angular parameters determine the positions of the suction cups on the circle, while the common radius determines their distance from the gripper Tool Center Point (TCP). For a fixed gripper height $z_0 = 0.326$ m, the Cartesian coordinates of the suction cups are obtained through the polar-to-Cartesian transformation

$$\mathbf{p}_i = \begin{bmatrix} r\cos(\alpha_i) \\ r\sin(\alpha_i) \\ z_0 \end{bmatrix}, \qquad i = 1,2,3. \tag{4.22}$$

The resulting Cartesian gripper configuration is written as

$$\mathbf{P} = [\mathbf{p}_1,\mathbf{p}_2,\mathbf{p}_3]. \tag{4.23}$$

Thus, the optimization is carried out in the reduced four-dimensional parameter space of $\mathbf{q}$, while the grasp evaluation operates on the derived Cartesian configuration $\mathbf{P}$.

This formulation reduces the dimensionality of the search space from six planar degrees of freedom,

$$[(p_{1,x},p_{1,y}),(p_{2,x},p_{2,y}),(p_{3,x},p_{3,y})] \in \mathbb{R}^6, \tag{4.24}$$

to four variables,

$$[r,\alpha_1,\alpha_2,\alpha_3]. \tag{4.25}$$

The reduction improves search efficiency and excludes many geometrically redundant or undesirable layouts while preserving sufficient flexibility for diverse triangular suction cup arrangements.

Assumption A triangular configuration of three suction cups is assumed to yield a stable grasp. This is motivated by the fact that three spatially distributed contact points form a statically stable support polygon, enabling effective load distribution and resistance to external disturbances.

4.2.2 Decision Variables and Bounds for Multi-Suction Cup Gripper

The optimization is performed over the reduced decision vector

$$\mathbf{q} = [r,\alpha_1,\alpha_2,\alpha_3]. \tag{4.26}$$

All variables are treated as continuous. The angular variables are bounded by

$$\alpha_i \in [0,2\pi), \qquad i = 1,2,3, \tag{4.27}$$

and the common radius is bounded by

$$r \in [r_{\min},r_{\max}]. \tag{4.28}$$

The radial bounds are determined separately for each workpiece. The workpiece is oriented such that its longest dimension lies in the horizontal plane, the horizontal bounding box is centred at $(x,y) = (0,0)$, and the base of the workpiece is placed at $z = 0$. This establishes a consistent coordinate system for defining the gripper-design bounds. Let $X_{\min}$, $X_{\max}$, $Y_{\min}$, and $Y_{\max}$ denote the coordinates of the bounding box of a workpiece in the x- and y-directions. The maximum radial distance is defined by the distance from the origin to the nearest bounding-box boundary:

$$r_{\max} = \min\left\{|X_{\min}|, |X_{\max}|, |Y_{\min}|, |Y_{\max}|\right\}. \tag{4.29}$$

The lower radial bound is derived from the suction-cup radius r_{cup} and is defined as

$$r_{\min} = \frac{2\sqrt{3}}{3}\, r_{\text{cup}}. \tag{4.30}$$

This lower bound corresponds to the circumradius of an equilateral triangle with side length $2r_{\text{cup}}$, i.e. the configuration in which neighbouring cups just touch.

All optimization methods generate candidate values within these bounds. In the evolutionary algorithms, the initial population is obtained by random sampling and subsequently modified through the operators of the selected algorithm. In Bayesian optimization, Ax first evaluates the center of the search space and then uses Sobol sampling to generate an initial space-filling design. After sufficient initialization data have been collected, Ax transitions to model-based Bayesian optimization.

4.2.3 Objective Function Evaluation of Multi-Suction Cup Gripper via API

The evaluation of objective values is decoupled from the optimization loop and delegated to an external evaluation service. For each candidate solution $\mathbf{q}$, the corresponding Cartesian suction cup coordinates are first computed by transforming the radial and angular parameters into 3D positions. The resulting Cartesian gripper configuration is then passed to the evaluation endpoint, which executes the grasp evaluation procedure and returns the objective values.

This process can be written as

$$\mathbf{P} = \Phi(\mathbf{q}), \qquad \mathbf{f}(\mathbf{q}) = \texttt{API_call}(\mathbf{P}) = \left[D,\, \hat{n}_{\mathrm{g}},\, \hat{A}\right], \tag{4.31}$$

where $\Phi(\cdot)$ denotes the transformation from the reduced parameterization to Cartesian suction cup coordinates, and the three objective components D, $\hat{n}_{\mathrm{g}}$ and $\hat{A}$ are defined in Section 4.2.3. Since every API call evaluates exactly one fixed configuration $\mathbf{P}$, the argument is omitted from the individual quality measures wherever it is unambiguous.

In the implementation, this transformation is carried out programmatically in the evaluation pipeline before grasp generation and objective computation. For a candidate radius and list of angles, the suction cup coordinates are constructed as

$$\mathbf{p}_i = (r\cos(\alpha_i),\, r\sin(\alpha_i),\, z_0)\,, \qquad i = 1,2,3. \tag{4.32}$$

For every candidate, the evaluation request performs the following steps:

1. Send the candidate decision vector to the external evaluation API.
2. Transform $[r,\alpha_1,\alpha_2,\alpha_3]$ into the Cartesian configuration $\mathbf{P} = \Phi(\mathbf{q})$.

3. Generate and filter the suction grasp candidates by seal formation and collision check.
4. Cluster the valid grasps and compute D, $\hat{n}_{\mathrm{g}}$ and $\hat{A}$.
5. Return the objective vector $\mathbf{f}(\mathbf{q})$ to the selected optimization framework.

This architecture enables:

- Separation of concerns between optimization logic and domain-specific evaluation
- Scalability through independent deployment of the evaluation service
- Reusability of the optimization framework for different problem domains

Grasp Evaluation

The evaluation service receives a Cartesian gripper configuration $\mathbf{P} = [\mathbf{p}_1,\mathbf{p}_2,\mathbf{p}_3]$ and computes the associated grasp quality measures through the following procedure.

1. **Candidate Grasp Generation and Feasibility Filtering**

   Suction grasp candidates are generated using a method developed by Fraunhofer IPA, following the automatic suction grasp generation procedure described by [68]. Points are uniformly sampled on the surface of the workpiece, and each sampled point defines a candidate grasp through its position and local surface normal. The local surface normal determines the approach direction of the suction cup.

   In the original method, each suction grasp candidate is tested for feasible seal formation using a quasi-static spring model of a suction cup, as introduced in [69]. If the grasp point is considered feasible, the candidate is further tested with a gripper collider model for collisions between the gripper and the object.

   In this thesis, the method was extended to a gripper with three suction cups. For each candidate gripper pose, the seal-formation test is performed for all three suction cup contact positions $\mathbf{p}_1$, $\mathbf{p}_2$, and $\mathbf{p}_3$. A candidate is considered feasible only if all three suction cups form a sufficient seal. Each feasible candidate is subsequently subjected to a geometric collision check: the gripper geometry is represented by a sampled point set, transformed into the workpiece frame by the candidate pose, and tested for proximity

against the workpiece point cloud. A candidate is rejected if any transformed gripper point lies closer than the clearance threshold $\delta_{\text{clr}} = 0.5\,\text{mm}$ to any workpiece point. The candidates that pass both the seal-formation and the collision checks form the set of valid grasps $\mathcal{H}_{\text{valid}}$, which is the input to the subsequent clustering step.

2. **Clustering of Grasps**

The set $\mathcal{H}_{\text{valid}}$ typically contains many grasps that are nearly identical in position and orientation. If the number of valid grasps $n_{\text{v}} = |\mathcal{H}_{\text{valid}}|$ exceeds the clustering budget $K_{\text{clst}} = 15$, the grasps are clustered using the greedy farthest-point sampling (FPS) algorithm of [95], which iteratively selects the grasp with the largest minimum distance to the already selected representatives; otherwise all valid grasps are retained.

Since grasp poses combine a translational and a rotational component, the sampling operates on a combined pose metric in which the Euclidean distance between the grasp positions $\mathbf{x}_i \in \mathbb{R}^3$ is normalised by the diagonal extent of the axis-aligned bounding box of all candidates, and the difference between the orientations $\mathbf{R}_i \in SO(3)$ is measured as the geodesic angle normalised by π [31]. With the resulting normalised terms $\hat{\rho}_{\text{pos}}$ and $\hat{\rho}_{\text{ori}}$ and an orientation weight $w \in [0,1]$,

$$\rho_w(i,j) = (1-w)\,\hat{\rho}_{\text{pos}}(i,j) + w\,\hat{\rho}_{\text{ori}}(i,j), \tag{4.33}$$

so that $w = 0$ corresponds to purely position-based and $w = 1$ to purely orientation-based sampling; here $w = 0.35$ is used. All distances are evaluated on the GPU, which keeps the selection tractable for large candidate sets.

The final number of representatives K^* is not predefined but obtained from an elbow criterion on the covering radius recorded during sampling: selection stops as soon as the relative coverage gain of an additional representative falls below $\tau = 0.05$, bounded from above by K_{clst}. The resulting subset $\mathcal{H}_{\text{clst}} \subseteq \mathcal{H}_{\text{valid}}$ with $|\mathcal{H}_{\text{clst}}| = K^*$ is passed on to the diversity evaluation.

3. **Diversity Metric Calculation**

A diversity score D is computed to quantify how broadly the resulting valid grasp set is distributed over the object, both in position and in orientation.

Let the set used for diversity evaluation contain n_g grasps. If clustering is performed, then

$$n_\mathrm{g} = |\mathcal{H}_\mathrm{clst}|, \tag{4.34}$$

otherwise

$$n_\mathrm{g} = |\mathcal{H}_\mathrm{valid}| = n_\mathrm{v}. \tag{4.35}$$

For $n_\mathrm{g} < 2$ no pairwise comparison is defined; in this case the corresponding component is set to zero.

The evaluated positions are the projections of the grasp points onto the mesh surface,

$$\mathbf{y}_i = \arg\min_{\boldsymbol{\xi}\in\partial\mathcal{W}} \|\boldsymbol{\xi} - \mathbf{x}_i\|_2, \qquad i = 1,\ldots,n_\mathrm{g}, \tag{4.36}$$

where $\partial\mathcal{W}$ denotes the surface of the workpiece mesh and $\mathbf{x}_i$ the original grasp point. The associated orientation matrices are $\{\mathbf{R}_1,\ldots,\mathbf{R}_{n_\mathrm{g}}\}$, with $\mathbf{R}_i \in SO(3)$.

To normalise the spatial terms with respect to object size, a characteristic object scale is defined as the Euclidean norm of the axis-aligned bounding-box extents,

$$s = \|\mathbf{e}\|_2, \tag{4.37}$$

where $\mathbf{e} \in \mathbb{R}^3$ contains the bounding-box side lengths.

The spatial diversity is assessed by two complementary quantities. The first is the normalised mean pairwise Euclidean distance between the grasp positions,

$$\begin{gathered} D_\mathrm{spread} = \frac{1}{s} \cdot \frac{1}{n_\mathrm{g}(n_\mathrm{g}-1)} \sum_{\substack{i,j=1 \\ i\neq j}}^{n_\mathrm{g}} \|\mathbf{y}_i - \mathbf{y}_j\|_2, \\ \mathbf{y}_i \in \partial\mathcal{W} \subset \mathbb{R}^3, \quad s = \|\mathbf{e}\|_2 \in \mathbb{R}_{>0}, \quad \mathbf{e} \in \mathbb{R}^3_{>0}, \quad n_\mathrm{g} \geq 2, \\ \|\mathbf{y}_i - \mathbf{y}_j\|_2 \leq s \implies D_\mathrm{spread} \in [0,1], \quad D_\mathrm{spread} := 0 \text{ for } n_\mathrm{g} < 2 \end{gathered} \tag{4.38}$$

Because a large mean distance can also be achieved by grasps concentrated on two opposite faces, a second term evaluates the angular coverage of the grasp positions as

seen from the object centroid $\mathbf{o}$. With the normalised view directions

$$\begin{gathered}\hat{\mathbf{u}}_i = \frac{\mathbf{y}_i - \mathbf{o}}{\|\mathbf{y}_i - \mathbf{o}\|_2 + \varepsilon}, \\ \mathbf{y}_i,\mathbf{o} \in \mathbb{R}^3, \quad \varepsilon \in \mathbb{R}_{>0},\, \varepsilon \ll 1, \quad \hat{\mathbf{u}}_i \in \mathbb{R}^3, \quad \|\hat{\mathbf{u}}_i\|_2 \in [0,1)\end{gathered} \tag{4.39}$$

where ε is a small constant guarding against division by zero, the pairwise cosine distance

$$\rho_{\cos}(i,j) = 1 - \hat{\mathbf{u}}_i^{\top}\hat{\mathbf{u}}_j \in [0,2] \tag{4.40}$$

vanishes for identical directions and attains its maximum for opposing directions. Its normalised mean defines the coverage term

$$D_{\text{cov}} = \operatorname{clip}\left(\frac{1}{2}\cdot\frac{1}{n_{\text{g}}(n_{\text{g}}-1)}\sum_{\substack{i,j=1\\ i\neq j}}^{n_{\text{g}}}\rho_{\cos}(i,j),\ 0,\ 1\right). \tag{4.41}$$

Both contributions are combined using the angular-coverage weight $\beta_{\text{cov}} = 0.5$:

$$\begin{gathered}D_{\text{pos}} = (1-\beta_{\text{cov}})\, D_{\text{spread}} + \beta_{\text{cov}}\, D_{\text{cov}}, \\ \beta_{\text{cov}} = 0.5 \in [0,1], \quad D_{\text{spread}},D_{\text{cov}} \in [0,1] \implies D_{\text{pos}} \in [0,1]\end{gathered} \tag{4.42}$$

so that a high spatial score requires the grasps to be both mutually distant and distributed around the object rather than clustered on a single side.

The orientation component is computed from the mean pairwise geodesic distance between rotation matrices. For two grasp orientations $\mathbf{R}_i, \mathbf{R}_j \in SO(3)$, the relative rotation is

$$\Delta\mathbf{R}_{ij} = \mathbf{R}_i^{\top}\mathbf{R}_j \tag{4.43}$$

and the associated geodesic distance is

$$\begin{gathered}\rho_{\text{geo}}(\mathbf{R}_i,\mathbf{R}_j) = \cos^{-1}\left(\frac{\operatorname{tr}(\Delta\mathbf{R}_{ij})-1}{2}\right), \\ \mathbf{R}_i,\mathbf{R}_j \in \mathrm{SO}(3), \quad \operatorname{tr}(\Delta\mathbf{R}_{ij}) \in [-1,3] \implies \tfrac{1}{2}(\operatorname{tr}(\Delta\mathbf{R}_{ij})-1) \in [-1,1], \\ \rho_{\text{geo}} : \mathrm{SO}(3)\times\mathrm{SO}(3) \to [0,\pi], \quad \rho_{\text{geo}}(\mathbf{R},\mathbf{R}) = 0\end{gathered} \tag{4.44}$$

where the argument of $\cos^{-1}$ is clipped to $[-1,1]$ for numerical stability. The normalised

orientation diversity follows as

$$D_{\text{rot}} = \frac{1}{\pi} \cdot \frac{2}{n_{\text{g}}(n_{\text{g}}-1)} \sum_{1 \le i < j \le n_{\text{g}}} \rho_{\text{geo}}(\mathbf{R}_i, \mathbf{R}_j). \tag{4.45}$$

The final diversity score weights the spatial component more strongly than the orientation component, since the orientation of suction grasps is largely determined by the local surface normal and therefore varies less independently. With $\lambda = 0.65$,

$$\begin{gathered} D = \text{clip}(\lambda\, D_{\text{pos}} + (1-\lambda)\, D_{\text{rot}},\ 0,\ 1)\,, \\ \lambda = 0.65 \in [0,1], \quad D_{\text{pos}}, D_{\text{rot}} \in [0,1] \implies \lambda D_{\text{pos}} + (1-\lambda) D_{\text{rot}} \in [0,1], \\ D \in [0,1] \text{ (the clip only compensates floating-point round-off)} \end{gathered} \tag{4.46}$$

where the final clipping prevents values marginally exceeding unity due to floating-point rounding. For large grasp sets the orientation term saturates around $D_{\text{rot}} \approx 0.7$, as uniformly distributed rotations do not reach the theoretical maximum of the normalised geodesic mean.

4. **Normalised Grasp Count**

Besides the spatial distribution of the grasps, their absolute number is relevant as well, since a layout that offers more valid grasps is more robust with respect to object pose and surrounding scene constraints. The evaluated grasp count is taken from the same set that is used for the diversity metric, i.e.

$$n_{\text{g}} = \begin{cases} |\mathcal{H}_{\text{clst}}|, & \text{if clustering is performed,} \\ |\mathcal{H}_{\text{valid}}|, & \text{otherwise.} \end{cases} \tag{4.47}$$

To keep this quantity comparable with the remaining objectives, it is expressed relative to the clustering budget K_{clst} (`max_cluster`), which is the maximum number of grasps retained by the clustering stage,

$$\hat{n}_{\text{g}} = \frac{n_{\text{g}}}{K_{\text{clst}}} \in [0,1]. \tag{4.48}$$

Since clustering is only applied when more than K_{clst} valid grasps exist, the condition $n_{\text{g}} \le K_{\text{clst}}$ holds in both cases and the ratio is bounded by unity without further clipping. A value of one indicates that the layout produced at least as many valid

grasps as the clustering budget permits, whereas a value of zero corresponds to a layout for which no valid grasp was found.

5. **Triangle Area Calculation**

   The area of the triangle formed by the three suction cups is computed from their Cartesian coordinates. With the planar coordinates $(p_{i,x},p_{i,y})$ of cup i,

$$\begin{aligned} A &= \frac{1}{2}\left|(p_{2,x}-p_{1,x})(p_{3,y}-p_{1,y})-(p_{3,x}-p_{1,x})(p_{2,y}-p_{1,y})\right|, \\ \mathbf{P} &\in \mathcal{P}_{\mathcal{W}}, \quad p_{i,x},p_{i,y} \in [-r_{\max}, r_{\max}], \quad A : \mathcal{P}_{\mathcal{W}} \to [0, A_{\max}], \\ A_{\max} &= \tfrac{3\sqrt{3}}{4} r_{\max}^2, \quad A = 0 \iff \text{the three cups are collinear} \end{aligned} \tag{4.49}$$

   To make this objective comparable in magnitude with the remaining objectives, the area is normalised by the largest triangle area that can be realised within the admissible mounting region. The suction-cup reference points are restricted to a circle of radius $r_{\max}$, and the maximum-area triangle inscribed in a circle is the equilateral one, so that

$$A_{\max} = \frac{3\sqrt{3}}{4}\, r_{\max}^2. \tag{4.50}$$

   The radius $r_{\max}$ is not a global constant but is determined per workpiece as the largest circle that still fits inside the admissible mounting bounds. Consequently $\hat{A}$ expresses the achieved area relative to what is geometrically possible for that specific workpiece.

   The normalised area objective is therefore

$$\hat{A} = \frac{A}{A_{\max}} \in [0,1]. \tag{4.51}$$

   This objective favours gripper layouts with a wider support polygon, which is associated with the assumption of improved grasp stability.

Continuity considerations Clustering is only applied when the number of valid grasps exceeds the clustering budget K_{clst} (`max_cluster`). If fewer valid grasps are available, clustering is skipped and the diversity metric is evaluated directly on the valid grasp set. Therefore, the omission of clustering does not by itself introduce a discontinuity in the objective function.

A discontinuity may arise only in the degenerate case where fewer than two valid grasps remain after collision filtering. Since both the spatial and orientation components of the diversity metric are based on pairwise distances, these terms cannot be evaluated meaningfully for $n_{\mathrm{v}} < 2$. In that case, the corresponding terms are set to zero. As a result, small changes in the gripper configuration may cause an abrupt transition between a non-zero diversity score and a degenerate zero-valued diversity contribution.

To reduce the likelihood of this case, the number of initially sampled surface points was increased from 100 to 1000. The denser surface sampling produces a larger set of candidate grasps and makes it much less likely that fewer than two valid grasps remain after filtering.

4.2.4 Objective Evaluation of Multi-Suction Cup Gripper

After grasp evaluation, each candidate layout is assigned the natural objective vector

$$\mathbf{f}(\mathbf{q}) = \left[D(\mathbf{P}),\, \hat{n}_{\mathrm{g}}(\mathbf{P}),\, \hat{A}(\mathbf{P})\right], \qquad \mathbf{P} = \Phi(\mathbf{q}), \tag{4.52}$$

where the ordering corresponds to the return order of the evaluation pipeline, so that $m = 3$ for this problem. In the remainder of the section the three components are written without arguments whenever the underlying layout is clear from the context.

All three components are maximised:

- D – the diversity score, where larger values indicate that the valid grasps are more broadly distributed over the object in position and orientation.
- $\hat{n}_{\mathrm{g}}$ – the normalised grasp count, where larger values indicate that the layout can grasp the object in more distinct ways and therefore offers greater robustness with respect to object pose and scene constraints.
- $\hat{A}$ – the normalised triangle area spanned by the three suction cups, where larger values correspond to a wider support polygon.

Following the convention of Section 4.1.3, the evolutionary algorithms receive

$$\mathbf{f}_{\mathrm{EA}}(\mathbf{q}) = -\,\mathbf{f}(\mathbf{q}) = \left[-D,\; -\,\hat{n}_{\mathrm{g}},\; -\,\hat{A}\right], \tag{4.53}$$

whereas the Bayesian implementation uses the unchanged observations

$$\mathbf{f}_{\mathrm{BO}}(\mathbf{q}) = \mathbf{f}(\mathbf{q}) = \left[D,\, \hat{n}_{\mathrm{g}},\, \hat{A}\right]. \tag{4.54}$$

The optimisation direction is specified through the Ax objective expressions, so the observations themselves are not negated before being reported. This preserves the original physical meaning of the objective values in the returned Pareto set.

4.2.5 Output of Multi-Suction Cup Gripper Optimization

In addition to the common output described in Section 4.1.8, the multi-suction cup optimization returns:

- The Pareto-optimal set $\mathcal{P}^*$ in the reduced decision space, with corresponding objective values
- The associated radial gripper parameterizations $\mathbf{q} = [r,\alpha_1,\alpha_2,\alpha_3]$

These solutions represent optimal trade-offs among grasp diversity, the number of valid grasps, and triangle area, corresponding to grasp versatility, robustness, and stability.

4.3 Implementation of Two-Finger Gripper Optimization

This section describes the problem-specific implementation for a two-finger gripper with a spline-based gripping contour. The common framework components are described in Section 4.1.

The evaluation service constructs the gripper geometry, voxelizes the gripper and workpieces, and calculates the grasp-quality objectives using a modified version of Snuggle-Pack [90]. The same decision variables, parameter bounds, geometry-generation pipeline, workpiece, orientations, and grasp-quality objectives are used for both optimization approaches. This enables their performance to be compared under a common evaluation budget and evaluation procedure.

4.3.1 Decision Variables and Bounds for Two-Finger Gripper Problem

A candidate gripper is represented by the decision vector introduced in Section 3.2.2,

$$\mathbf{q} = \begin{bmatrix} \ell_{\mathrm{gn}} & h_1 & h_2 & h_3 \end{bmatrix}^{\mathrm{T}}, \qquad d = 4, \tag{4.55}$$

where the variables are transmitted to the evaluation API in the order

$$\mathbf{q} = [\texttt{length_gn}, \texttt{h_p1_spline}, \texttt{h_p2_spline}, \texttt{h_p3_spline}]^{\mathrm{T}}. \tag{4.56}$$

The decision variables and their bounds are summarized in Table 4.4.

Table 4.4: Decision variables of the two-finger gripper optimization problem.

Variable	**Configuration parameter**	**Lower bound**	**Upper bound**
ℓ_{gn}	`length_gn`	20	50
h_1	`h_p1_spline`	2	10
h_2	`h_p2_spline`	2	10
h_3	`h_p3_spline`	2	10

All four decision variables are treated as continuous. Candidate values generated by either optimization framework are restricted to the bounded search space

$$\mathcal{Q} = [20{,}50] \times [2{,}10]^3 \subset \mathbb{R}^4. \tag{4.57}$$

For evolutionary optimization, the variables are initially sampled within these bounds and subsequently modified by the operators of the selected algorithm. For Bayesian optimization, Ax initially evaluates the center of the search space and generates additional initialization candidates using Sobol sampling. After sufficient observations have been collected, Ax transitions to model-based candidate generation.

4.3.2 Objective Function Evaluation of Two-Finger Gripper via API

For every candidate, the decision variables are sent to the evaluation API as an ordered list. The API inserts these values into a complete gripper configuration, from which the jaw

geometry is generated:

$$\mathbf{q} \longrightarrow \mathbf{c}_{\mathrm{gr}}(\mathbf{q}) \longrightarrow \mathcal{G}(\mathbf{q}), \tag{4.58}$$

where $\mathbf{c}_{\mathrm{gr}}(\cdot)$ denotes the mapping to the complete gripper configuration and $\mathcal{G}(\cdot)$ the resulting jaw geometry.

Besides the decision variables, the configuration $\mathbf{c}_{\mathrm{gr}}(\cdot)$ contains fixed parameters, including the contour type, the basic jaw dimensions (depth, width, and length), the jaw angle, the stroke, the axial offset of the contour, and the tool-center-point translation.

For every candidate, the evaluation request performs the following steps:

1. Send the candidate decision vector to the external evaluation API.
2. Map the decision variables to the complete gripper configuration $\mathbf{c}_{\mathrm{gr}}(\mathbf{q})$.
3. Generate the spline-based gripper geometry using CadQuery.
4. Voxelize the gripper and the configured workpiece in all predefined orientations.
5. Calculate the FFT-based collision maps using the modified Snuggle-Pack procedure.
6. Calculate one proximity KPI for each predefined orientation of the configured workpiece.
7. Return the KPI vector to the selected optimization framework.

Parametrization of the Jaw Contour

The contact contour of the rigid jaw is described by an interpolating spline in the contour plane, which is spanned by the jaw depth direction and the jaw length direction. The decision vector consists of the contour length ℓ_{gn} and three depth parameters h_1, h_2, and h_3. These four values define the five control points of the spline,

$$\mathbf{s}_0 = (0,0), \qquad \mathbf{s}_j = \left(h_j,\, \tfrac{j}{4}\,\ell_{\mathrm{gn}}\right), \quad j = 1,2,3, \qquad \mathbf{s}_4 = (0,\ell_{\mathrm{gn}}), \tag{4.59}$$

so that the control points are distributed at the relative contour positions 0, 0.25, 0.50, 0.75, and 1.0 along the contour length. The first and last control point are pinned to zero depth, which guarantees that the contour begins and ends flush with the planar jaw face and that

the resulting profile is a closed recess rather than an open step. The intermediate depths h_1, h_2, and h_3 control how far the contour intrudes into the jaw at the quarter, half, and three-quarter positions and therefore determine the shape of the form fit. The contour curve is the interpolating spline

$$\begin{gathered}\mathcal{S}(\mathbf{q}) = \operatorname{spline}\left(\mathbf{s}_0,\ldots,\mathbf{s}_4\right),\\ \mathbf{s}_j \in \mathbb{R}^2,\, j = 0,\ldots,4, \quad \mathcal{S}(\mathbf{q}) : [0,1] \to \mathbb{R}^2, \quad \mathcal{S}(\mathbf{q}) \in C^1\Big([0,1],\mathbb{R}^2\Big),\\ \exists\, 0 = u_0 < u_1 < \cdots < u_4 = 1 :\ \mathcal{S}(\mathbf{q})(u_j) = \mathbf{s}_j \text{ (parameter values not necessarily } j/4)\end{gathered} \tag{4.60}$$

which passes through all five control points.

The spline is closed into a planar profile by straight segments up to a fixed cutting depth and is then extruded through the full jaw depth to remove the enclosed material from the basic jaw body. The geometry is constructed with CadQuery [96] and exported to the mesh representation required for evaluation. Since only ℓ_{gn} and the three depth parameters are varied, every candidate geometry differs from the others exclusively in the contact contour, while all remaining jaw dimensions remain identical.

Each candidate geometry $\mathcal{G}(\mathbf{q})$ is generated and voxelized only once per evaluation request and is subsequently reused for all configured grasps, so that the cost of the geometry generation is independent of the number of objectives.

Grasp Set Definition

A grasp specifies the pose in which the workpiece is presented to the jaws. In the evaluation pipeline, the gripper remains stationary in its nominal pose, and the workpiece is reoriented for each grasp.

Only the orientation component of the generated grasp poses is used, since the relative gripper–workpiece position is determined by the collision- and proximity-based placement search described in Section 4.3.2. A grasp γ_i is therefore fully described by a spatial rotation $\mathbf{R}_i \in \mathrm{SO}(3)$ applied to the workpiece,

$$\mathcal{W}_i = \mathbf{R}_i\,\mathcal{W}, \qquad i = 1,\ldots,n_{\mathrm{or}}, \tag{4.61}$$

where $\mathcal{W}$ denotes the workpiece in its reference orientation. The translation components of the generated poses are discarded. The grasp set is

$$\mathcal{R} = \{\mathbf{R}_1, \ldots, \mathbf{R}_{n_{\mathrm{or}}}\}, \qquad n_{\mathrm{or}} = 10. \tag{4.62}$$

In the present study, $n_{\mathrm{or}} = 10$ orientations are configured. Because the retained poses originate from the feasibility filtering and the farthest-point clustering of the generation procedure, the orientations are neither restricted to a single rotation axis nor obtained as independent samples of a parametric distribution. They are instead a spatially spread subset of the physically admissible grasps of the given workpiece, which ensures that the evaluated orientations are mutually dissimilar and at the same time realizable on the part geometry.

The grasp set is generated once before the optimization starts and is stored as part of the workpiece list. It is therefore identical for all candidates and for all repetitions of a run, which makes the objective values of different candidates directly comparable. The orientations are fixed quantities and are not optimization variables.

Each optimization run is defined for exactly one workpiece together with its corresponding predefined grasp set. The workpiece geometry and the grasp set $\mathcal{R}$ are constant throughout a run, and all objectives of a run refer to the same workpiece evaluated in its n_{or} predefined orientations. Different workpieces are treated as separate optimization problems and are optimized in separate runs, since each workpiece requires its own task-specific jaw contour.

Grasp Evaluation

The generated gripper is evaluated separately for every grasp γ_i, $i = 1, \ldots, n_{\mathrm{or}}$, of the workpiece configured for the run. The gripper remains in its assembly pose, while the workpiece is rotated into the orientation of the respective grasp. Both geometries are voxelized into three-dimensional occupancy grids. For the gripper, the surface voxelization of each part is closed by morphological dilation and hole filling. Every voxel carries the label of the part it belongs to, so the two jaws can be addressed individually.

The evaluation builds on a modified version of Snuggle-Pack [90], a three-dimensional packing approach that combines FFT-based spatial analysis with proximity heuristics. Instead of

placing objects inside a container, the same principle is used to search for the relative placement at which both jaws enclose the workpiece as closely as possible without penetrating it.

For each jaw $j \in \{\mathrm{L},\mathrm{R}\}$ with occupancy grid $\chi_{\mathcal{J}_j}$, the overlap with the workpiece grid $\chi_{\mathcal{W}_i}$ is computed for all discrete displacements $\mathbf{v}$ in a single transform step:

$$\zeta_{i,j} = \mathrm{DFT}^{-1}\Big(\mathrm{DFT}(\chi_{\mathcal{J}_j})\ \mathrm{DFT}(\chi_{\mathcal{W}_i})\Big), \qquad \zeta_{i,j}(\mathbf{v}) < \varepsilon_{\mathrm{col}} \iff \mathbf{v} \text{ is collision-free.} \tag{4.63}$$

The threshold $\varepsilon_{\mathrm{col}}$ absorbs the round-off of the single-precision FFT. Because the grids are padded only to a common size, the transform is cyclic. Only the jaws enter the collision test while the gripper base is not considered.

The overlap map of each jaw is then convolved with a one-sided Gaussian proximity kernel that points along the closing direction of that jaw. The resulting proximity map is high for displacements at which workpiece material lies directly in front of the jaw face, that is, where a small further closing motion would establish contact. The kernel's standard deviations are given in millimetres and converted to voxel units via the voxel edge length. The kernel acts only in the plane orthogonal to the gripper length axis, so in practice only the width and height deviations shape the score. Colliding displacements are excluded from the proximity map.

The placement search follows the kinematics of a parallel gripper. For every position in the length–height plane, each jaw independently selects its best collision-free position along the closing axis. The two jaw proximities are then summed and maximized over that plane, which yields the proximity score p_i. Both jaws therefore share a common placement in length and height but move independently along the closing direction. All operations run as GPU-accelerated PyTorch FFT convolutions. The score is dimensionless, unnormalized and Gaussian-weighted. It increases the more closely both jaws enclose the workpiece, but it is neither a distance nor a voxel count, and its magnitude depends on the workpiece and on the voxel resolution.

The difference between the two selected jaw positions along the closing axis defines a signed opening width δ_i. A negative value means the jaws close onto the workpiece. A non-negative value means the placement would require the jaws to open beyond their modelled position. Only the sign is evaluated; the magnitude is not compared against the stroke. The KPI

reported to the optimization framework is

$$k_i(\mathbf{q}) = \begin{cases} p_i(\mathbf{q}), & \delta_i(\mathbf{q}) < 0, \\ 0, & \delta_i(\mathbf{q}) \geq 0, \end{cases} \qquad i = 1, \ldots, n_{\text{or}}. \tag{4.64}$$

Feasibility is assessed per grasp, so an infeasible orientation penalizes only its own objective. Since collision-free proximity values are non-negative, zero is the worst attainable outcome. An infeasible orientation is therefore treated as a worst-case objective value rather than as a hard constraint.

4.3.3 Objective Evaluation of Two-Finger Gripper

The evaluation service returns one grasp-quality KPI per configured grasp. Because a single workpiece and its associated grasp set are used per optimization run, the number of objectives is determined directly by the size of that grasp set:

$$m = n_{\text{or}}. \tag{4.65}$$

The individual KPIs are not aggregated into a scalar value; each grasp retains its own objective, which preserves the information about which orientations a given contour favours.

Since a larger proximity KPI represents a better grasp result, the natural optimization problem is

$$\begin{gathered} \max_{\mathbf{q} \in \mathcal{Q}} k_i(\mathbf{q}), \qquad i = 1, \ldots, m, \\ \mathcal{Q} \subset \mathbb{R}^4 \,, \quad m = n_{\text{or}} = 10, \\ k_i : \mathcal{Q} \to \mathcal{K} \subseteq \mathbb{R}_{\geq 0} \end{gathered} \tag{4.66}$$

with the objective vector

$$\mathbf{f}(\mathbf{q}) = \begin{bmatrix} k_1(\mathbf{q}) & k_2(\mathbf{q}) & \cdots & k_m(\mathbf{q}) \end{bmatrix}^{\mathrm{T}} \in \mathbb{R}^m_{\geq 0}. \tag{4.67}$$

Because the four decision variables define a single jaw contour that is shared by all grasps, the objectives are generally conflicting: a contour whose depth profile conforms closely to the workpiece in one orientation does not necessarily conform in the remaining ones. The

optimization therefore yields Pareto-optimal contours that represent different compromises in geometric conformity across the ten predefined orientations.

Following the convention of Section 4.1.3, the evolutionary algorithms receive the negated KPI vector

$$\mathbf{f}_{\mathrm{EA}}(\mathbf{q}) = -\,\mathbf{f}(\mathbf{q}) = \begin{bmatrix} -k_1(\mathbf{q}) & -k_2(\mathbf{q}) & \cdots & -k_m(\mathbf{q}) \end{bmatrix}^{\mathrm{T}}, \tag{4.68}$$

whereas the Bayesian implementation uses the unchanged observations

$$\mathbf{f}_{\mathrm{BO}}(\mathbf{q}) = \mathbf{f}(\mathbf{q}). \tag{4.69}$$

Consequently, the observed KPI values do not need to be negated before they are attached to the corresponding Ax trial, which preserves the physical meaning of the KPI values when the final Pareto frontier is returned. For performance comparisons, the results of both optimization frameworks are converted to the common natural maximization convention $\mathbf{f}(\mathbf{q})$, so that all reported objective values correspond to the ten grasp-quality KPIs in their original direction and magnitude.

4.3.4 Output of Two-Finger Gripper Optimization

In addition to the common output described in Section 4.1.8, the two-finger optimization returns:

- The Pareto-optimal decision vectors $\mathbf{q} = [\ell_{\mathrm{gn}}, h_1, h_2, h_3]^{\mathrm{T}}$
- The corresponding objective KPI values $\mathbf{f}(\mathbf{q})$

The non-dominated solutions therefore represent the best trade-offs between grasp quality across the predefined orientations of the configured workpiece.

5 Validation

5.1 Validation of Results for Suction-Cup Gripper

5.2 Validation of Results for Two-Finger Gripper

6 Evaluation

6.1 Results for Suction-Cup Gripper

6.1.1 Results for Evolutionary Algorithms

6.1.2 Results for Bayesian Algorithms

6.2 Results for Two-Finger Gripper

6.2.1 Results for Evolutionary Algorithms

6.2.2 Results for Bayesian Algorithms

7 Summary

7.1 Conclustion

7.2 Limitations

7.3 Future Work

A Appendix

Alles was den Hauptteil unnï¿½tig vergrï¿½ï¿½ert hï¿½tte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

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